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N706B

Hardware User Guide

Cat.1 Module
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This document is specifically for N706B.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

This document is applicable to N706B series.

It defines the features, indicators, and test standards of the N706B module and provides reference for the hardware design of each interface.

Audience

This document is intended for system engineers (SEs), development engineers, and test engineers.

Change History

Issue	Date	Change	Changed By
1.0	2024-12	Initial release	Liang Hui
1.1	2025-01	• Updated N706B-CN power consumption parameters • Updated module weight • Updated UART baud rate information	Liang Hui
1.2	2025-03	• Updated recommended power supply design diagram • Updated description of USIM pull-up resistor	Wang Haoyu
1.3	2025-05	• Updated power consumption parameters for N706B-EA • Updated power consumption parameters for N706B-LA • Updated RF characteristics	Wang Haoyu
1.4	2025-05	Updated power voltage drop diagram	Wang Haoyu

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Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

Related Documents

Neoway_N706B_Datasheet

Neoway_N706B_Product_Specifications

Neoway_N706B_AT_Command_Mannual

Neoway_N706B_EVK_User_Guide

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1 Safety Recommendations

Ensure that this product is used in compliance with the requirements of the country and the environment. Please read the following safety recommendations to avoid body hurts or damages of product or workplace:

- Do not use this product at any places with a risk of fire or explosion such as gasoline stations, oil refineries, and so on.

  If the product is used in a place with flammable gas or dust such as propane gas, gasoline, or flammable spray, the product will cause an explosion or fire.

- Do not use this product in environments such as hospital or airplane where it might interfere with other electronic equipment.

  If the product is used in medical institutions or on airplanes, electromagnetic waves emitted by this product may interfere with surrounding equipment.

Follow the requirements below in design and use of the application for this module:

- Do not disassemble the module without permission from Neoway. Otherwise, we are entitled to refuse to provide further warranty.
- Design your application correctly by referring to the HW design guide document and our review feedback on your PCB design. Connect the product to a stable power supply and lay out traces following fire safety standards.
- Please avoid touching the pins of the module directly in case of damages caused by ESD.
- Do not insert/remove a SIM card or memory card into/from the module while it is not powered off.

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2 About N706B

N706B is an LTE industrial-grade cellular module developed based on the ASR 1605 platform, providing connectivity on FDD-LTE and TDD-LTE networks. With dimensions of (17.7±0.20) mm × (15.8±0.20) mm × (2.3±0.15) mm, the module features a variety of hardware interfaces, supporting functions such as UART, I2C, Wi-Fi Scan, and ADC. It is suitable for developing IoT communication devices like wireless meter reading terminals, handheld POS systems, and industrial routers.

N706B has the following features:

- ARM Cortex-R5 processor, 614 MHz CPU clock speed, 32 KB I-cache and 32 KB D-cache.
- Supported network mode: LTE Cat.1
- Supported functional interfaces: USIM, UART, I2C, USB2.0, SPI, and ADC.

2.1 Product Overview

The following lists the frequency bands and memory information that N706B supports.

Table 2-1 Supported frequency bands

Model	Region	Band	GNSS	WIFI_SCAN	Memory
N706B-CN-10	Chinese mainland, India	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B39, B40, B41	Supported	Supported	RAM: 32 Mb ROM: 32 Mb
N706B-CN-11	Chinese mainland, India	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B39, B40, B41	Not supported	Supported	RAM: 32 Mb ROM: 32 Mb
N706B-CN-21	Chinese mainland, India	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B39, B40, B41	Not supported	Not supported	RAM: 32 Mb ROM: 16 Mb
N706B-EA-11	Europe	FDD-LTE: B1, B3 B5, B7, B8, B20, B28	Not supported	Supported	RAM: 32 Mb ROM: 32 Mb
N706B-LA-11	Latin America	FDD-LTE: B1, B2, B3	Not supported	Supported	RAM: 32 Mb ROM: 32 Mb

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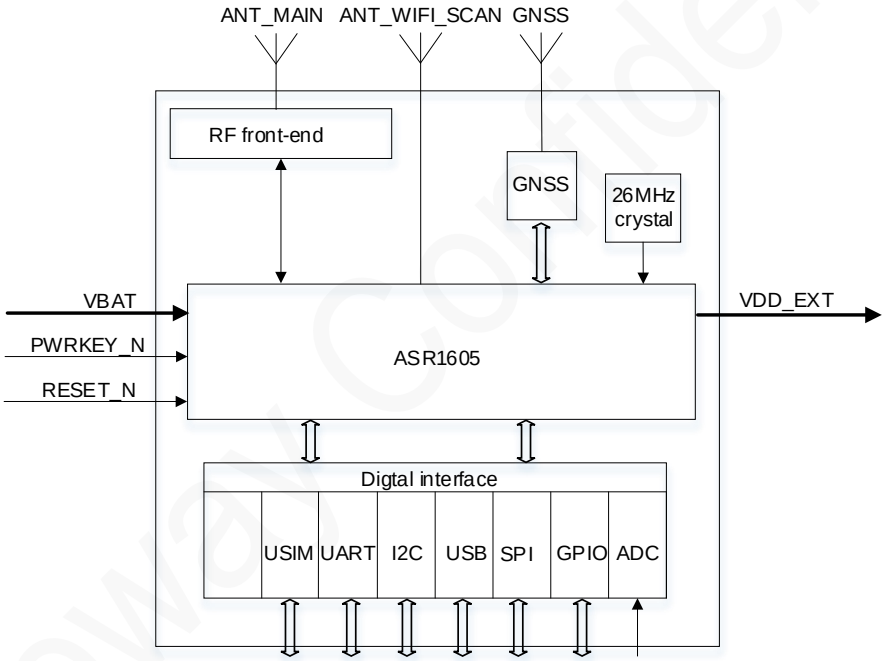
	B4, B5, B7, B8, B28, B66

2.2 Block Diagram

N706B includes the following functional units:

- Platform chip unit
- Digital interfaces (USIM, UART, I2C, USB 2.0, SPI)
- Analog interface (ADC)

Figure 2-1 Block diagram



2.3 Basic Features

Features	Description
Physical features	Dimensions: (17.7 ± 0.20) mm × (15.8 ± 0.20) mm × (2.3 ± 0.15) mm Package: N706B-CN: LCC+LGA (109 pins) N706B-EA&LA: LGA (109 pins) Weight: about 1.4g
Temperature	Operating: -30°C to +75°C

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ranges	Extended[1)]: -40°C ~ +85°C Storage: -40°C to +90°C
Operating voltage	VBAT: 3.4 V to 4.3 V, typical value: 3.8 V
Operating current	Sleep mode[2)]: < 1.5mA Standby mode[3)]: < 14mA Operating mode[4)]: < 700mA
Application processor	ARM Cortex- R5 processor, main frequency 614 MHz
Band	See Table 2-1.
Wireless rate	FDD-LTE: Cat 1, Max 10 Mbps (DL)/Max 5 Mbps (UL) TDD-LTE: Cat 1, Max 8 Mbps (DL)/Max 2 Mbps (UL)
Application Interfaces	4G antenna, WiFi_SCAN, GNSS antenna. The characteristic impedance of each antenna is 50 Ω.
	Three UART interfaces with a default baud rate of 115200 bps. MAIN_UART for AT communications. AUX_UART is used for module data transmission (unavailable in sub-models with GNSS). DBG_UART for AP-side log capturing.
	2 USIM interfaces: SIM1 supports 1.8V/3V auto-sensing, while SIM2 only supports 1.8V.
	One USB 2.0 interface, supporting only device mode.
	One SPI interface, supporting SPI Flash and both master/slave modes.
	There are two 12-bit ADC interfaces with a voltage detection range of 0.1V to 1.2V.
	Two I2C interfaces, for master mode only.
AT commands	3GPP Release 13

Extended[1] In the event that the temperature in the operating environment surpasses the designated thresholds, certain RF performance indicators of the module may deteriorate and fall short of the 3GPP specification requirements. However, despite this, the module's regular functionality will not be significantly affected. It is possible for the RF performance to return to the required 3GPP standards once the temperature is normalized.

Current in sleep mode[2]: means the current drawn by the module in sleep mode, a low power consumption state, in which its RF function is functioning properly but its peripheral interfaces are disabled. If there is an incoming call or SMS, the module will exit from the sleep mode, and after the incoming call or voice instant messaging has ended, the module will re-enter the sleep mode.

Current in standby mode[3]: means the current drawn by the module in a normal operating mode, but no data service is being processed.

Current in operating mode[4]: means the current drawn by the module when there is data communications.

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	Neoway extended AT commands
SMS	PDU, TXT
Data	PPP, RNDIS, ECM
Protocols	TCP, UDP, MQTT, FTP, HTTP/HTTPS, SSL, TLS
Certification approval	CCC*, SRRC*, RoHS*, CTA*

* means under development

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3 Reference Standard

N706B is designed by referring to the following standards:

- 3GPP TS 36.521-1 V13.0.0 User Equipment (UE) conformance specification; Radio transmission and reception; Part 1: Conformance Testing
- 3GPP TS 21.111 V13.0.0 USIM and IC card requirements
- 3GPP TS 51.011 V4.15.0 Specification of the Subscriber Identity Module -Mobile Equipment (SIM-ME) interface
- 3GPP TS 31.102 V13.0.0 Characteristics of the Universal Subscriber Identity Module (USIM) application
- 3GPP TS 31.111 V13.0.0 Universal Subscriber Identity Module (USIM) Application Toolkit (USAT)
- 3GPP TS 27.007 V13.0.0 AT command set for User Equipment (UE)
- 3GPP TS 27.005 V13.0.0 Use of Data Terminal Equipment – Data Circuit terminating Equipment (DTE - DCE) interface for Short Message Service (SMS) and Cell Broadcast Service (CBS)

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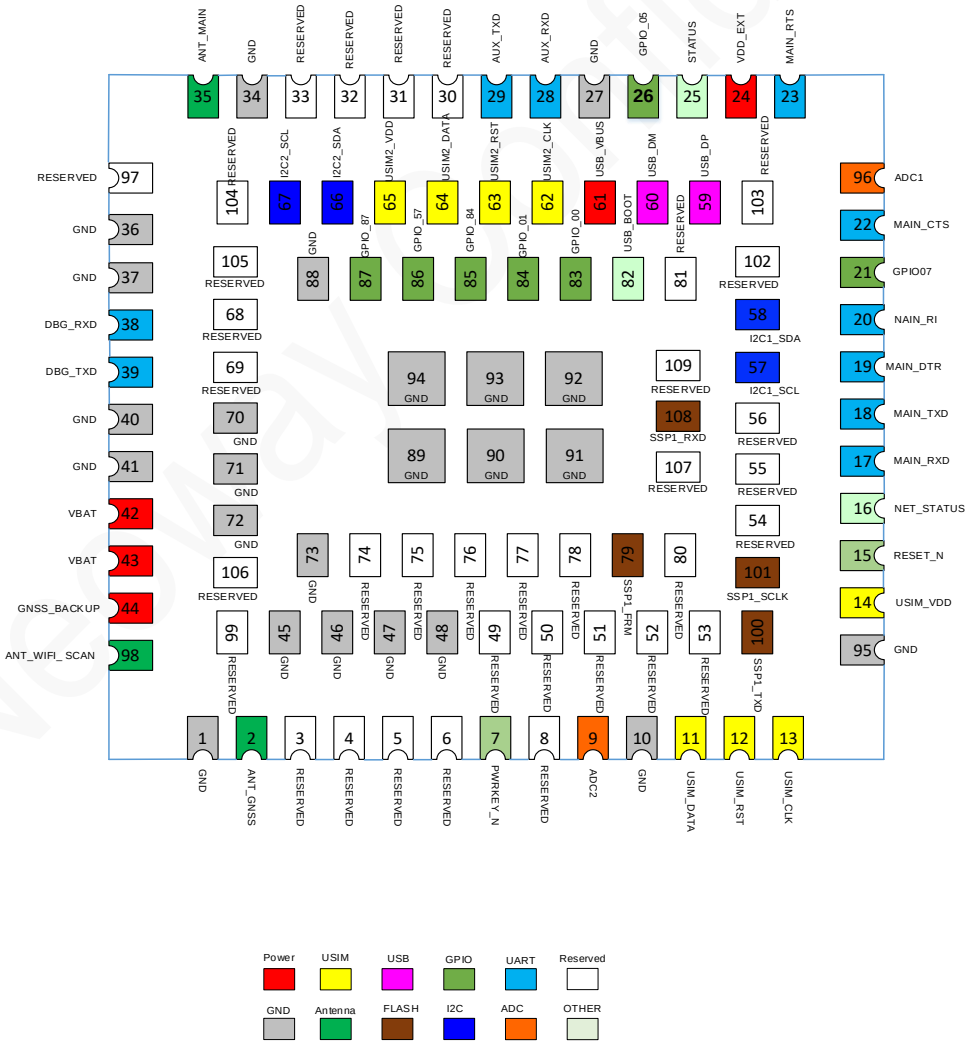
4 Pad Layout

The N706B modules are designed with a total of 109 pads, which are distributed across a 109-pin LGA + 109-pin LGA package.

4.1 Pad Layout

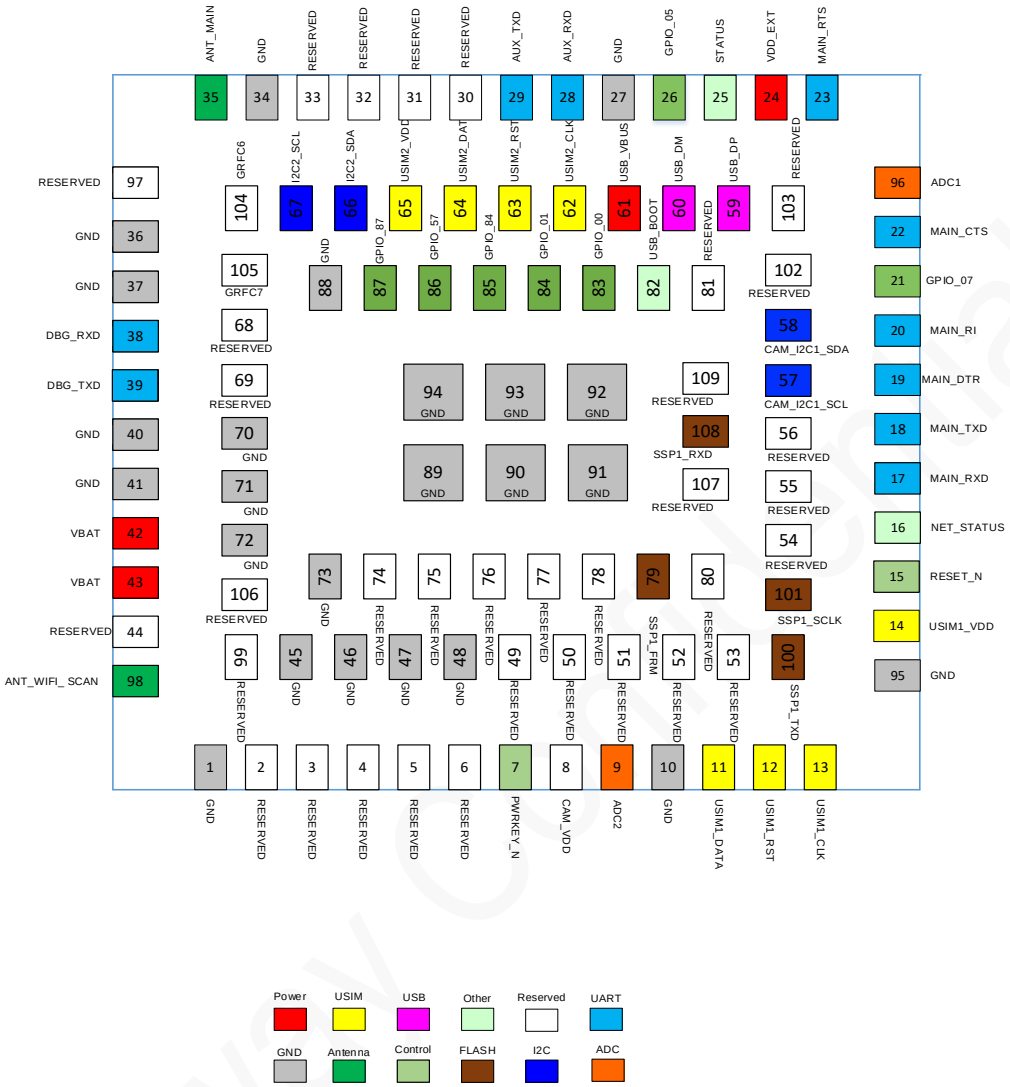
The following figure shows the pad layout of N706B.

Figure 4-1 N706B-CN pad layout (top view)



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Figure 4-2 N706B-EA/LA pad layout (top view)


ANT_MAIN
GND
RESERVED
RESERVED
RESERVED
RESERVED
AUX_TXD
AUX_RXD
GND
GPIO_05
STATUS
VDD_EXT
MAIN_RTS
35 34 33 32 31 30 29 28 27 26 25 24 23
GRFC6
I2C2_SCL
I2C2_SDA
USIM2_VDD
USIM2_DAT
USIM2_RST
USIM2_CLK
USB_VBUS
USB_DM
USB_DP
RESERVED
104 67 66 65 64 63 62 61 60 59 103
GND
GPIO_87
GPIO_57
GPIO_84
GPIO_01
GPIO_00
USB_BOOT
RESERVED
105 88 87 86 85 84 83 82 81 102
GRFC7
RESERVED
RESERVED
RESERVED
RESERVED

PO	Power output		
PI	Power input		
AO	Analog output		
AI	Analog input		
AIO	Analog input and output		
DC characteristics			
Interface type	Power domain	Logic level characteristics	
USIM	P1: Both 1.8 V and 3 V SIM types are supported.	1.8 V DC characteristics: V_{IH}=0.7 x V_{DD_P1}~ V_{DD_P1} V_{IL}=0V~0.3 x V_{DD_P1} V_{OH}=0.9 x V_{DD_P1}~ V_{DD_P1} V_{OL}=0V~0.1 x V_{DD_P1}	3 V DC characteristics: V_{IH}=0.7 x V_{DD_P1}~ V_{DD_P1} V_{IL}=0V~0.3 x V_{DD_P1} V_{OH}=0.9 x V_{DD_P1}~ V_{DD_P1} V_{OL}=0V~0.1 x V_{DD_P1}
GPIO	P2:1.8V	V_{IH}=0.7 x V_{DD_P2}~ V_{DD_P2} V_{IL}=0V~0.3 x V_{DD_P2} V_{OH}=0.9 x V_{DD_P2}~ V_{DD_P2} V_{OL}=0V~0.1 x V_{DD_P2}	

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Table 4-2 Pin Description

Signal	Pin SN	I/O type	Functional description	DC characteristics	Default pull[5]	Remarks
Power interfaces						
VBAT	42, 43	PI	Main power supply input	V_{min}=3.4 V V_{norm}=3.8 V V_{max}=4.3 V	/	The external power supply must ensure at least 2 A current.
VDD_EXT	24	PO	1.8 V power output	V_{norm}=1.8 V I_{max}=20 mA	/	IO interface, only used for voltage-level translation. Leave this pin open if unused.
GNSS_BACKUP	44	PI	1.8V input	V_{norm}=1.8 V I_{max}= 100uA	/	For GNSS backup power, the version without GNSS can be left unconnected.
GND	1, 10, 27, 34, 36, 37, 40, 41, 45, 46, 47, 48, 70, 71, 72, 73, 88, 89, 90, 91, 92, 93, 94, 95				/	Make sure all GND pins are grounded.
Control interfaces						
PWRKEY_N	7	DI	Module on/off control	-	/	Active low For details, see section 5.2.1 .
RESET_N	15	DI	Module reset input	-	/	-
MAIN_DTR	19	DI	Sleep wakeup pin	-	DOWN	-

[5] The module's Default pull refers to the state before the module's software is fully operational, right after the module powers on and starts up.

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UART interface						
MAIN_RXD	17	DI	Data input	P2	UP	Used for AT communications
MAIN_TXD	18	DO	Data output	P2	UP	
MAIN_CTS	22	DI	UART clear to send	P2	UP	-
MAIN_RTS	23	DO	UART request to send	P2	UP	Do not repurpose as GPIO. Leave this pin open if unused.
MAIN_DTR	19	DI	Sleep/ wakeup pin	P2	DOWN	-
MAIN_RI	20	DO	RING pin	P2	DOWN	-
AUX_RXD	28	DI	UART data input	P2	DOWN	AUX_UART is used for data transmission (unavailable in sub-models with GNSS).
AUX_TXD	29	DO	UART data output	P2	DOWN	
DBG_RXD	38	DI	UART data input	P2	UP	DBG_UART is only used for Debug. Can be used to capture AP log.
DBG_TXD	39	DO	UART data output	P2	UP	
USIM interface						
USIM1_DATA	11	B	USIM data	P1	/	Connecting a 4.7 kΩ pull-up resistor to USIM_VCC is required.
USIM1_RST	12	DO	USIM reset	P1	/	-
USIM1_CLK	13	DO	USIM clock	P1	/	-
USIM1_VDD	14	PO	USIM power supply	P1	/	1.8V/3V self-adaptive
USIM2_DATA	64	B	USIM data	P2	DOWN	Connecting a 4.7 kΩ pull-up resistor to USIM_VCC is required.

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USIM2_RST	63	DO	USIM reset	P2	DOWN	-
USIM2_CLK	62	DO	USIM clock	P2	DOWN	-
USIM2_VDD	65	PO	USIM power supply	P2	/	Only support 1.8V.
USB interface						
USB_VBUS	61	PI	USB power supply	3.0 V - 5.25 V, typical value: 5 V	/	-
USB_DP	59	AIO	USB data +	-	/	Route the DM and DP traces as differential pairs with 90 Ω impedance.
USB_DM	60	AIO	USB data -	-	/	
ADC interface						
ADC2	9	AI	General A/D interface	-	/	12-bit, detectable voltage range: 0.1 V - 1.2V. Leave this pin open if unused.
ADC1	96	AI	General A/D interface	-	/	12-bit, detectable voltage range: 0.1 V - 1.2V. Leave this pin open if unused.
I2C interface						
I2C1_SCL	57	DO	I2C clock	P2	UP	Connect this pin to VDD_EXT through a pull-up 4.7kΩ resistor.
I2C1_SDA	58	B	I2C data	P2	UP	
I2C2_SDA	66	B	I2C data	P2	UP	
I2C2_SCL	67	DO	I2C clock	P2	UP	

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SPI interface*						
SSP1_FRM	79	DO	SPI chip select	P2	DOWN	Support external SPI FLASH.
SSP1_SCLK	101	DO	SPI clock	P2	DOWN	
SSP1_TXD	100	B	SPI output	P2	UP	
SSP1_RXD	108	B	SPI input	P2	UP	
Antenna interface						
ANT_WIFI_SCAN	98	AI	WIFI_SCAN antenna	-	/	50 Ω impedance.
ANT_MAIN	35	AIO	Main antenna	-	/	50 Ω impedance.
ANT_GNSS	2	AIO	GNSS antenna	-	/	50 Ω impedance.
GPIO						
GPIO_00	83	DO/DI		P2	DOWN	-
GPIO_01	84	DO/DI		P2	DOWN	
GPIO_07	21	DO/DI		P2	UP	
GPIO_05	26	DO/DI		P2	DOWN	
GPIO_84	85	DO/DI		P2	DOWN	
GPIO_57	86	DO/DI		P2	DOWN	
GPIO_87	87	DO/DI		P2	UP	
Other functional interfaces						
STATUS	25	DO	Operating status indication	P2	DOWN	Leave this pin open if unused.
NET_STATUS	16	DO	Network status indication	P2	DOWN	Leave this pin open if unused.

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USB_BOOT	82	DI	Emergency download control	P2	/	Ground this pin and then power up the module to enter the USB emergency download mode.
RESERVED	3 - 6, 8, 30 - 33, 49 - 56, 68, 69, 74 - 78, 80, 81, 97, 99, 104, 105, 102 - 106, 107, 109	-	Reserved pin	-	/	Reserved for future function expansion or functions not open to the user. There may be more than one pin named RESERVED, which may have different functions or definitions. Leave these RESERVED pins open.

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5 Application Interfaces

N706B encompasses a versatile suite of interfaces, designed to cater to a wide spectrum of application scenarios. Ranging from control and communications, to peripherals and other functions, these interfaces offer comprehensive solutions tailored to specific needs.

This chapter describes how to design each interface and provides reference designs and guidelines.

5.1 Power Interfaces

The power interfaces play a pivotal role in application design, with both the schematic design and PCB layout serving as critical components. Their configuration directly impacts the performance of customers' applications. Please read the design guidelines of power supply and comply with the correct design principles to obtain the optimal circuit performance.

Signal	Pin SN	I/O	Functional description	Remarks
VBAT	42, 43	PI	Main power supply of the module	V_{min}=3.4 V V_{norm}=3.8 V V_{max}=4.3 V
VDD_EXT	24	PO	1.8 V power output	V_{norm}=1.8 V I_{max}=20 mA
GNSS_BACKUP	44	PI	For GNSS backup power, the version without GNSS can be left unconnected.	V_{min}=1.65 V V_{norm}= 1.8 V V_{max}=3.6 V I_{max}= 100 uA
GND	1, 10, 27, 34, 36, 37, 40, 41, 45, 46, 47, 48, 70, 71, 72, 73, 88, 89, 90, 91, 92, 93, 94, 95			Make sure all GND pins are grounded.

5.1.1 VBAT

The power supply design encompasses two key facets: schematic design and PCB layout. These are integral in sustaining the optimal functionality and safety of the module.

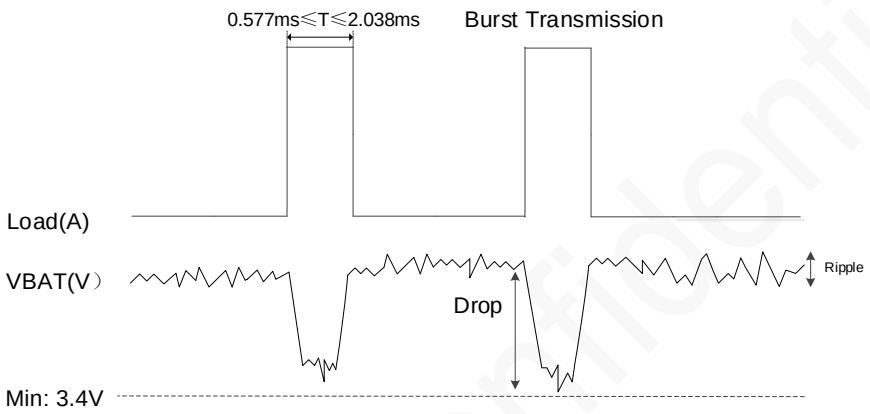
Schematic Design of Power Supply

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The module may experience a momentary peak current of up to 2 A during network registration. It is necessary to parallel a large capacitor on VBAT to enhance the current sustaining ability. This is to ensure that the VBAT input voltage does not fall below the module's operating voltage during these moments of peak current.

Never bridge a voltage drop between a higher input and the module power supply with a diode. The forward voltage drop Vf of a diode has two characteristics: one is that it increases with the increase of forward current; the other is that it significantly increases at low temperatures. If there is instantaneous high current, the above characteristics may cause the module working voltage to be unstable, or even damage the module.

Figure 5-1 Voltage drop of the power supply

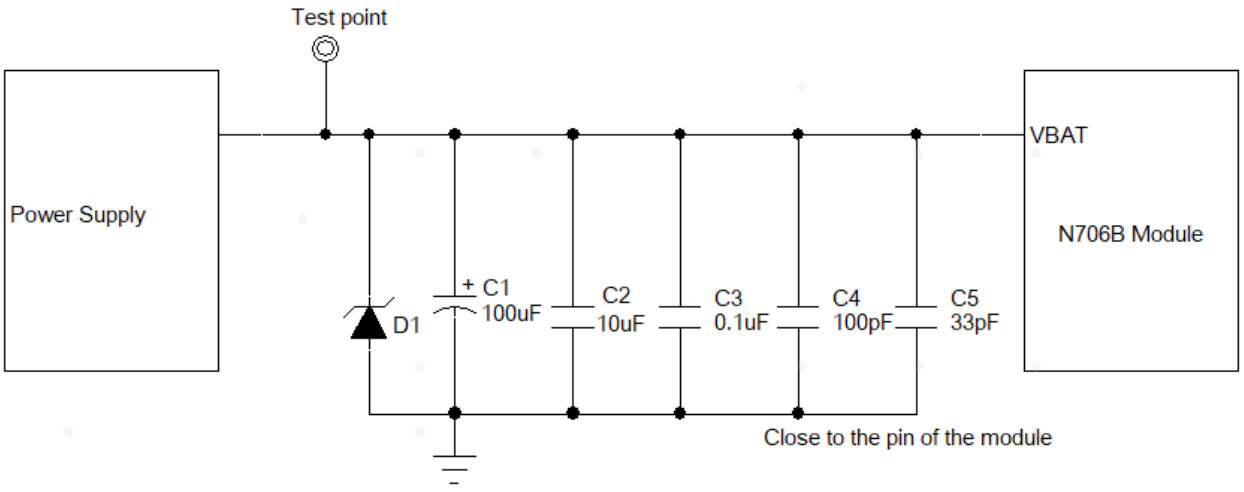


The power supply design of the N706B module depends on the input voltage, categorized into:

- 3.4 V - 4.3 V power input (typical value: 3.8 V, battery supplied)
- 4.3 V - 5.5 V power input (typical value: 5.0 V, internal computer rectifier)
- 5.5 V - 24 V power input (typical value: 12 V, system DC power in industries like automotive)

Design Recommendations for 3.4 V - 4.3 V Input:

Figure 5-2 Recommended power supply design 1



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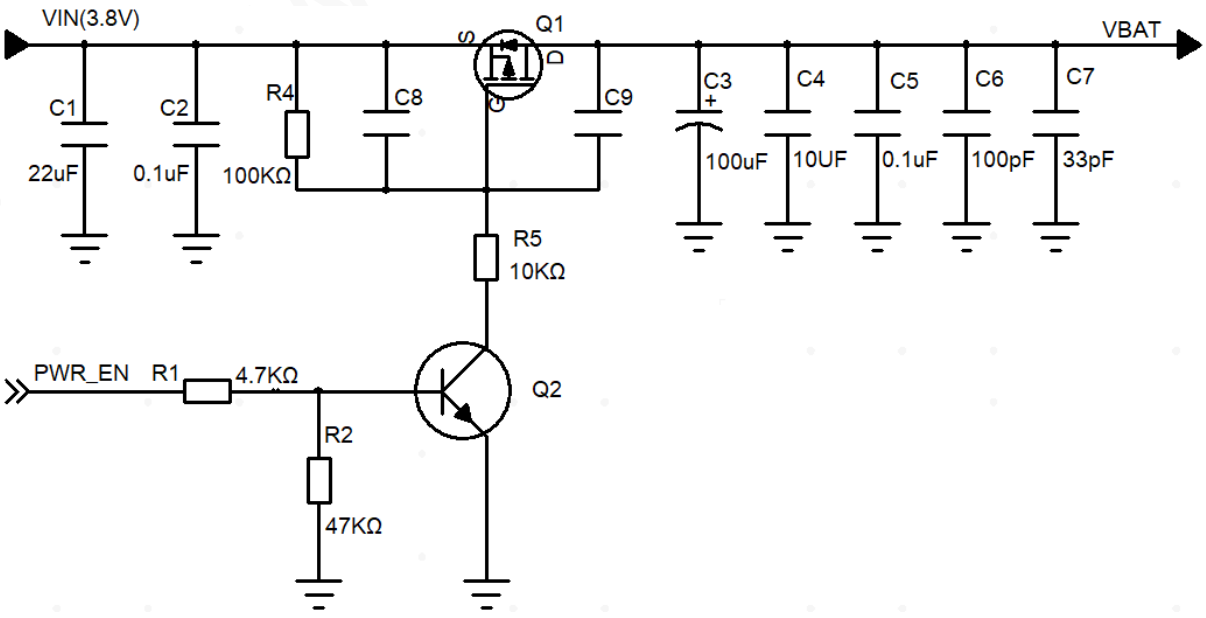
- The maximum input voltage of the module power supply is 4.3 V, and the typical value is 3.8 V.
- The TVS diode D1 should have a reverse working voltage VRWM of 4.5V, peak power Ppp of 400W (tp=8/20μS), and a clamping voltage not exceeding 10V. Keep the placement of the TVS diode close to the power input interface to suppress surge voltage before it enters back-end circuits to prevent the back-end component and the module from damaging.
- Use a large bypass tantalum capacitor (220 μF or 100 μF) or an aluminum capacitor (470 μF or 1000 μF) at C1 to decrease voltage drops during bursts. Its maximum safe operating voltage should be larger than 2 times the voltage of the power supply.
- Keep the placement of the low-ESR bypass capacitors (C2, C3, C4, C5) with capacitance not greater than 0.7 Ω close to the module to filter out high-frequency jamming from the power supply.

PCB Design Guidelines

- When connecting an external power supply to the module, the VBAT trace width should be no less than 2 mm. In principle, the longer the VBAT trace, the wider it should be. Consider both VBAT and return GND paths when routing power traces.
- Ensure the trace passes through the TVS and large capacitors before reaching the module's power pin, with smaller capacitors placed closer to the VBAT pin of the module.
- Pins 40 and 41 of the module are for connecting to the power supply's GND, and these connections should be as short and direct as possible.
- All GND pins of the module should be connected to the main ground with as many vias as possible to ensure the integrity of the board's grounding.

If it is necessary to control the power supply, the following circuit design is recommended:

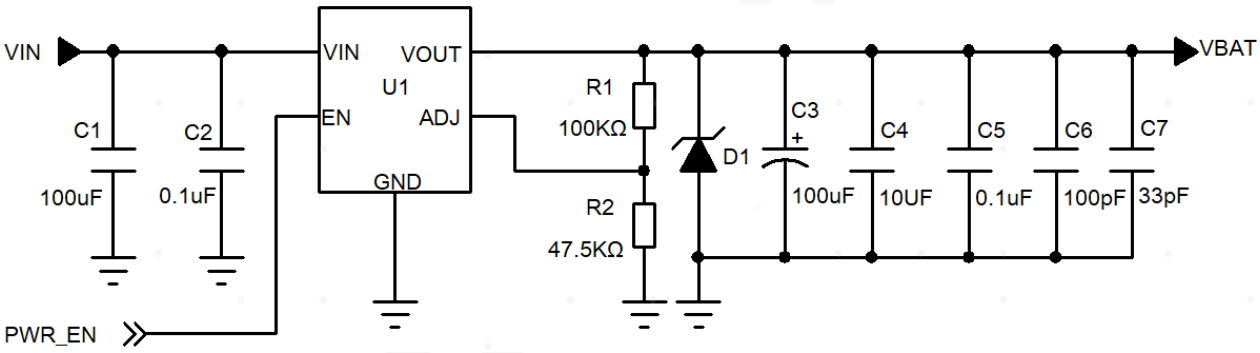
Figure 5-3 Recommended power supply design 2



- Utilize an enhanced p-MOSFET at Q1, of which the safe operating voltage is at least 12 V (V_{dss}=-12 V) and drain current is at least 3.5 A ($I_{D(MAX)}$=-3.5 A) and Rds is low ($R_{ds(on)}$ =108 mΩ).
- Select a common NPN transistor at Q2. Reserve enough tolerances for R1 and R2, ensuring that R2 is at least 10 times greater than R1, as the break-over voltage on the transistor's base increases at low temperatures.
- Keep the placement of C3 close to the module. Select a tantalum electrolytic capacitor with large capacitance (220 µF or 100 µF) or an aluminum electrolytic capacitor (470 µF or 1000 µF) at C3 to improve the instantaneous large freewheeling current of the power supply, and its withstand voltage value should be larger than 2 times the power supply voltage.
- Keep the placement of the low-ESR bypass capacitors (C4, C5, C6, C7) with capacitance not greater than 0.7 Ω close to the module to filter out high-frequency jamming from the power supply.

The design recommendations for the 4.3 V - 5.5 V power input are as follows:

Figure 5-4 Recommended power supply design 3



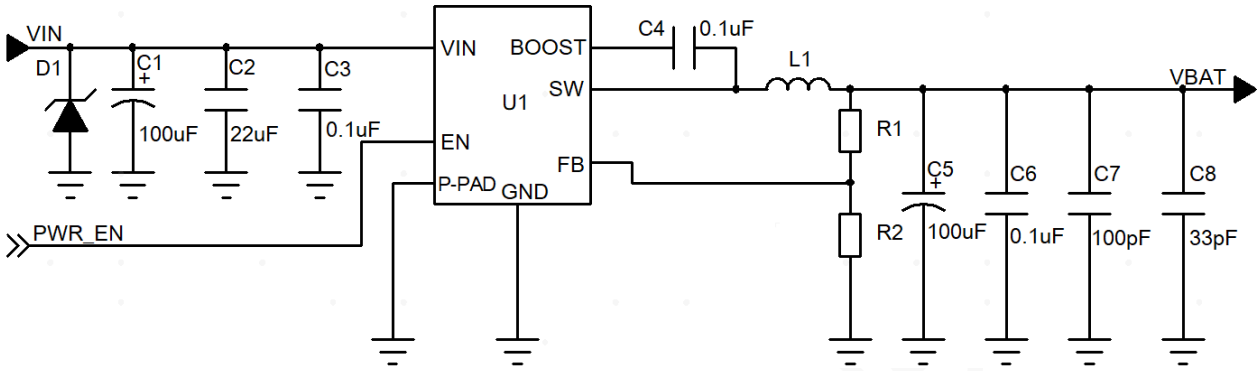
- Design with LDO is simpler and more efficient when the output of power supply is close to the permissible voltage across VBAT.
- Select an LDO with maximum output current above 2 A at U1 to ensure the normal performance of the module.
- In order to get a stable power source, it is recommended to use a TVS diode with suggested low reverse voltage (V_{RWM} = 4.5 V) and peak power (Ppp = 2800 W (tp = 8/20 us)) at D3. Keep the placement of the TVS diode close to the power input interface to suppress surge voltage before it enters back-end circuits to prevent the back-end component and the module from damaging.
- Keep the placement of C3 close to the module. Select a large tantalum electrolytic capacitor (220 µF or 100 µF) or aluminum electrolytic capacitor (470 µF or 1000 µF) at C3 to improve the instantaneous large freewheeling current of the power supply, and its withstand voltage value should be larger than 2 times the power supply voltage.

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- Keep the placement of the low-ESR bypass capacitors (C4, C5, C6, C7) with capacitance not greater than 0.7 Ω close to the module to filter out high-frequency jamming from the power supply.

The design recommendations for the 5.5 V - 24 V power input are as follows:

Figure 5-5 Recommended power supply design 4



- If there is a high voltage drop between the power input and VBAT, a DC-DC design should be selected for higher efficiency, and the output current should be at least 2 A.
- Keep the TVS diode (D1) close to the power input interface to ensure that the power surge voltage is clamped before entering the back-end circuit, thus protecting the back-end components and modules.
- Keep the placement of C5 close to the module. Select a large tantalum electrolytic capacitor (220 μF or 100 μF) or aluminum electrolytic capacitor (470 μF or 1000 μF) at C3 to improve the instantaneous large freewheeling current of the power supply, and its withstand voltage value should be larger than 2 times the power supply voltage.
- Keep the low ESR (≤ 0.7 Ω) bypass capacitors (C6, C7, C8) as close to the module as possible to filter out high-frequency interference in the power supply.

When the module fails to be switched off normally due to abnormality, it is recommended to cut off the power first to shut down the module, and then power on again to restart the module.

PCB Layout Guidelines

Place an ESR capacitor at the output of the power source to suppress the peak current. Place TVS diodes at the input end of the module to suppress voltage spikes and protect back-end devices. Several key points in power supply design are summarized below:

- The TVS diode can absorb instantaneous high-power pulse, and withstand instantaneous pulse current peaks up to tens or even hundreds of amperes, with extremely short response time of voltage clamping. Place the TVS diode near the power input. This helps control surge voltage and prevents pulses from affecting nearby PCB wires.

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- Place bypass capacitors as close as possible to the power supply interface of the module to filter out high-frequency noise signals in the power supply.
- For the main power circuit of the module, ensure that the PCB trace is wide enough that 2 A current can be safely passed, with no significant loop voltage drop. Keep PCB trace width be at least 2 mm and ensure that the ground plane of the power supply part is as complete as possible. Try to make the power traces short and wide.
- Keep noise-sensitive circuits, such as audio circuits and RF circuits far away from the power circuitry, especially when the DC-DC power supply is used.
- The voltage frequency of the SW pin of the DC-DC power supply is high, so the loop area should be minimized. Keep sensitive components far away from the SW pin of the DC-DC component to prevent noise coupling. Place feedback components as close as possible to the FB pin.
- The thermal-dissipation pads and GND pads of the DC-DC chip should be grounded to ensure good thermal dissipation and noise signal isolation.

5.1.2 VDD_EXT



It is recommended to use VDD_EXT only for voltage-level translation and an ESD protector should be added.

N706B provides one VDD_EXT interface that supports 1.8 V voltage and up to 20 mA current.

5.1.3 GNSS_BACKUP

Pin #44 can be optionally used as the GNSS backup power pin. If not used, it should be left floating. The voltage range is 1.65V to 3.6V, with a typical value of 1.8V, and an external current supply of 100μA is required. A 1μF filter capacitor should be placed close to the module pin. For more details, please contact Neoway FAE.

PCB Layout Guidelines

- Ensure ESD protection when using the pin.
- The hot start time refers to the time taken by the GNSS to restart when the module is powered on.
- The VBACKUP pin is used for the GNSS hot start function, supplying power to the GNSS unit's RTC and backup RAM. When the GNSS main power is cut off (controlled internally by the module), the module can retain key ephemeris and almanac data for the hot start functionality.

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5.2 Control Interfaces

Signal	Pin SN	I/O	Functional description	Remarks
RESET_N	15	DI	Module reset input	Active low
PWRKEY_N	7	DI	Module on/off control	Active low

5.2.1 Module Power-on

When designing the power on/off circuit, the control circuit must strictly follow our recommended design guidelines. The RC components cannot be removed. Refer to recommended circuits Figure 5-6 and Figure 5-7 for guidance.

Table 5-1 Power-on description

Notice Power-on trigger method	Supported or not	Power-off method	Notice
Button	Supported	Press the power-on button for greater than 3s and then release it to switch off the module.	For example: Do not disconnect the power supply directly when the module is in power-on mode, as this may damage the internal flash of the module.
Pulse	Supported	A low pulse input (controlled by MCU) for greater than 3s can shut down the module.	
Auto power-on	Supported	Send the power-off AT command to the module and then disconnect its power source.	If the module is powered off using AT commands but the system does not lose power, the module may unexpectedly power back on after shutdown, which could potentially lead to damage to the module's internal flash.

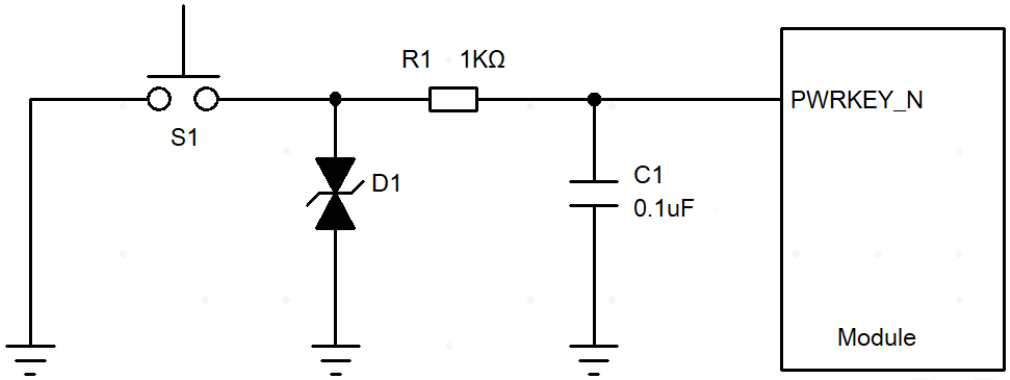
Power-on via PWRKEY_N

The following show two recommended reference designs of module startup:

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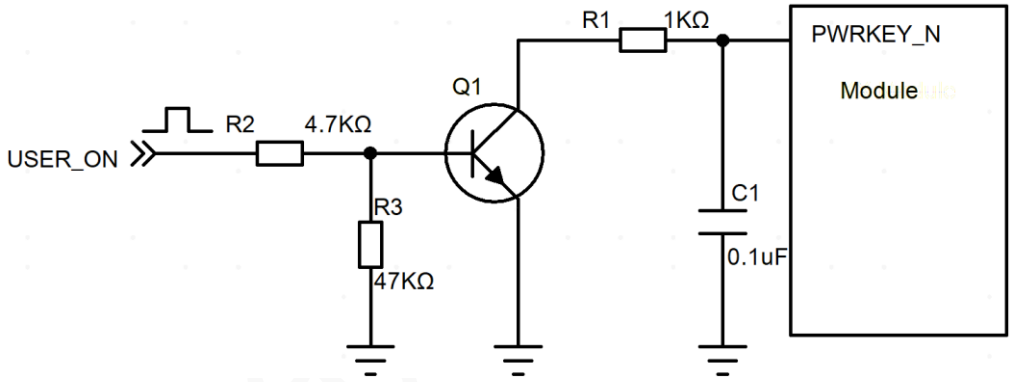
- Figure 5-6 shows the reference design for push-button startup.

Figure 5-6 Reference design for push-button startup



- Figure 5-7 shows the reference design of module power-on through an MCU

Figure 5-7 Reference design for MCU-controlled startup



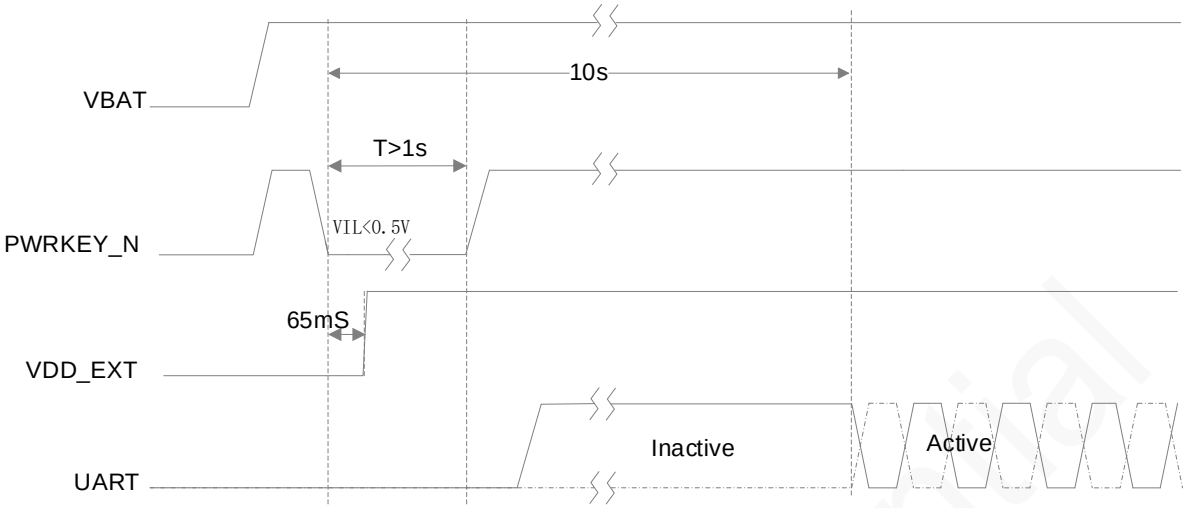
Power-on Process

After VBAT is powered on, releasing the PWRKEY_N pin after inputting a low-level pulse of more than 1s can trigger the module to start up, and it can be controlled by AT commands normally after 10s. It is important not to connect an external resistor with high resistance in series with the PWRKEY_N pin, as this could keep the pin at a high level and prevent the module from starting. It is recommended to connect a 1 kΩ resistor in series to the PWRKEY_N pin.

- Avoid performing operations on the module until initialization is complete, as the states of each pin are uncertain if the initialization process has not been finished. Refer to Figure 5-8 for the power-on process.

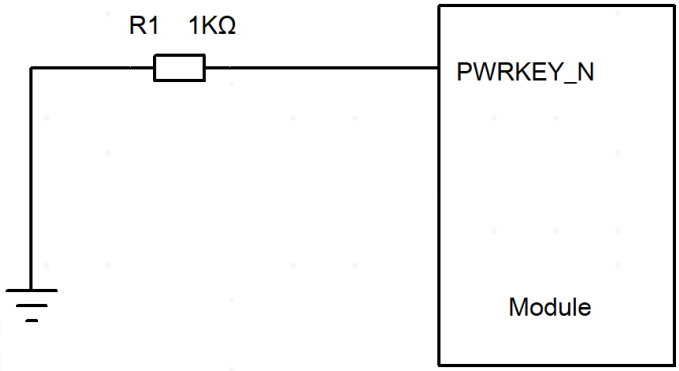
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Figure 5-8 Power-on process



Auto Power-on

Figure 5-9 Reference design of automatic power-on once powered-up



To implement automatic startup with no need for shutdown, you can pull PWRKEY_N down to ground. A pull-down resistance of 1 kΩ is recommended.

5.2.2 Module Power-off

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Do not disconnect the power supply when the module is on; flash memory may be damaged. Do not directly disconnect the power of the module in power-on mode, if there is not any abnormality.

When shutting down using an AT command, ensure that PWRKEY_N is not in a low-level state after executing the shutdown command. If it is, the module will restart after completing the shutdown process without the power being disconnected.

Power-off Methods:

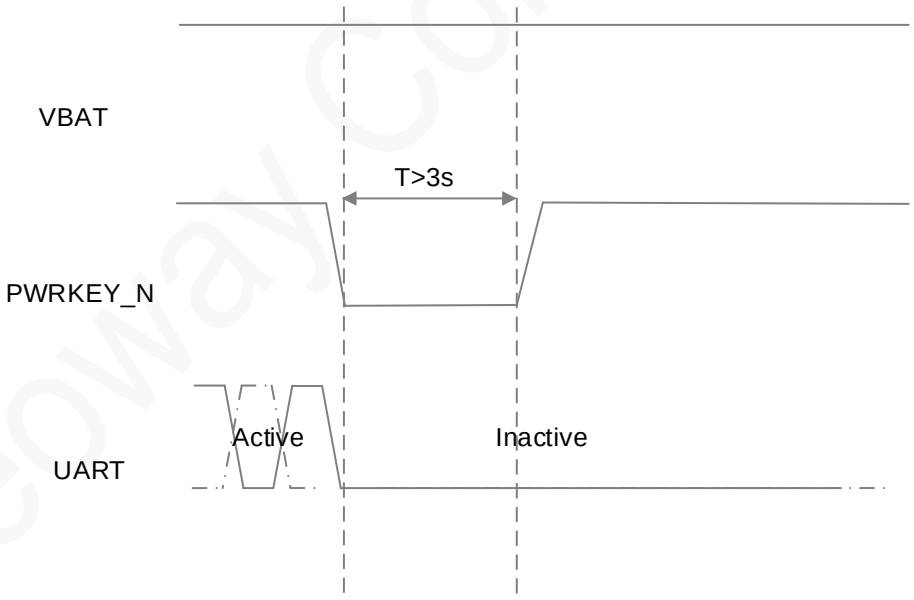
- Hardware power-off via PWRKEY_N
- Software Power-off via AT command

Hardware Power-off via PWRKEY_N

When the module is in power-on mode, to turn off the module, the PWRKEY_N pin must be asserted low for greater than 3s and then released.

The following figure shows the hardware power-off process:

Figure 5-10 N706B hardware power-off process



Software Power-off via AT Command

When the module is in power-on mode, to turn off the module, send it the power-off AT command. Whether the module is switched off by hardware or software method, their mechanisms are the same, that is, by applying a low level at the PWRKEY_N pin to switch off the module.

For details, see the *Neoway_N706B_AT_Commands_Manual*.

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5.2.3 Resetting Methods

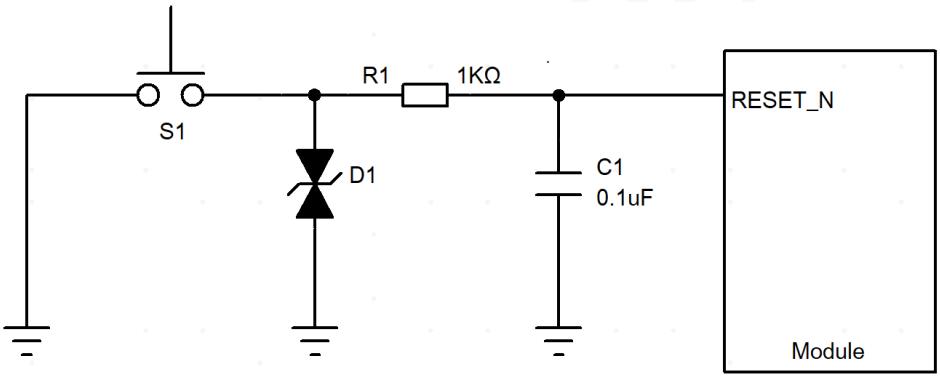
If the module encounters an abnormal condition (e.g., system crash), it is recommended to first attempt recovery using software AT commands and the PWRKEY_N shutdown. If both methods fail, use the RESET_N pin to perform a system reset to ensure normal operation. Therefore, it is recommended to include a RESET_N control circuit in your design. The control circuit must strictly adhere to our recommended design and must not omit the RC components. Refer to the recommended circuits Figure 5-11 and Figure 5-12 for guidance.

In normal operation, a low-level pulse exceeding 50 ms applied to the RESET_N pin triggers a module reset. The reset process is detailed as shown in Figure 5-11.

The following show two recommended reference designs of resetting N706B:

- Reference design for push-button reset

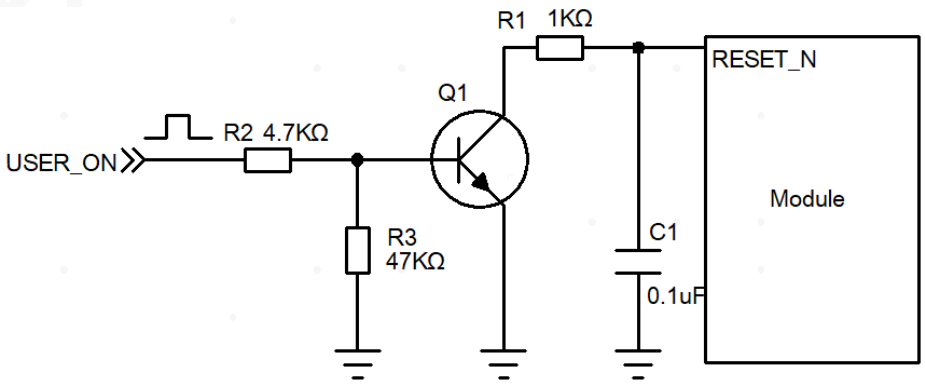
Figure 5-11 Reference design for push-button reset



- Reference design of MCU-controlled reset

  It is recommended to add a triode to for isolation. Refer to the following figure for specific design.

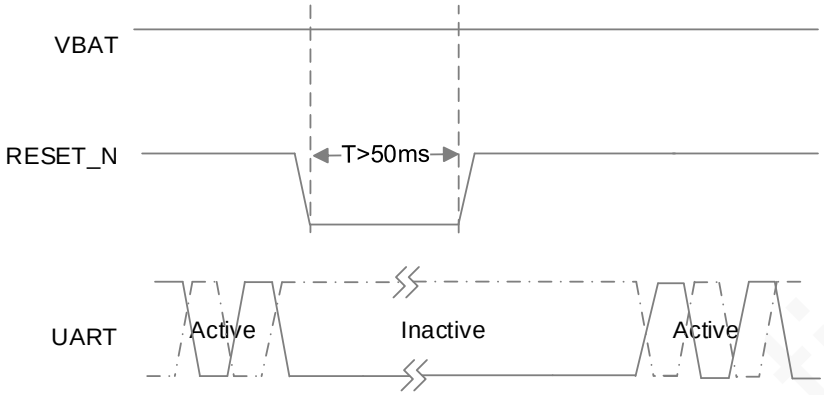
Figure 5-12 Reference design for reset circuit in which a triode is used for isolation



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N706B module reset process is shown in the following figure.

Figure 5-13 Module reset process



Design considerations

- Ensure that the PWRKEY_N and RESET_N pins do not have large load capacitors (maximum load capacitance should not exceed 10 nF).
- The RESET_N signal is sensitive to interference; therefore, PCB traces for this signal should be kept as short as possible and surrounded by ground to minimize noise.
- The module RESET_N reset pin provides a soft reset without power loss, which only resets the baseband and does not reset the PMU. It is important to note that a non-power-loss reset may not reset the module in all scenarios. It is recommended that users use a power-loss reset, which requires coordination with the module's power control circuitry to ensure proper module operation.
- If the electromagnetic environment is complex, it is advisable to use a dual transistor level conversion circuit to isolate and protect the RESET_N pin.

5.2.4 DTR

Signal	Pin SN	I/O	Functional description	Remarks
MAIN_DTR	19	DI	Sleep and wake-up	Requires software adaptation and supports edge-triggered wake-up.

The MAIN_DTR pin is the module's control pin for entering and exiting sleep mode. It must be used in conjunction with AT commands. The module currently supports entering sleep mode via USB or MAIN_UART by sending the appropriate AT commands. For details, see *Neoway_N706B_AT_Commands_Manual*.

The following show the processes:

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Figure 5-14 Process of entering into sleep mode

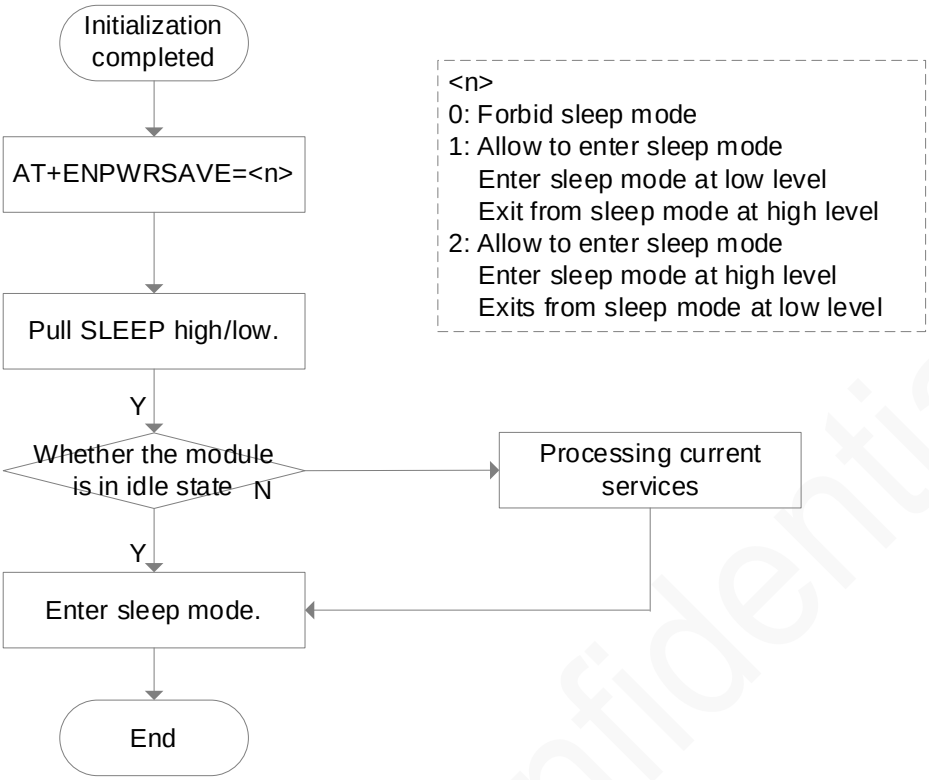
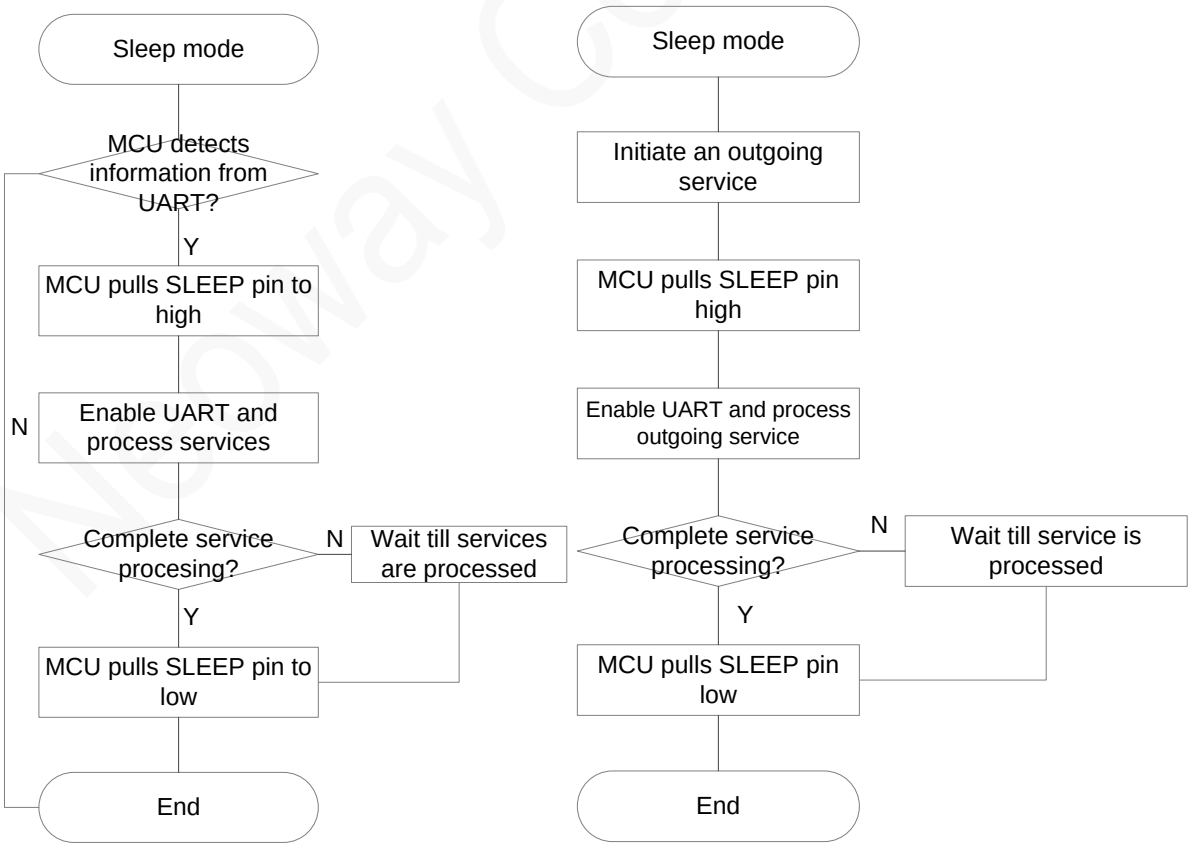
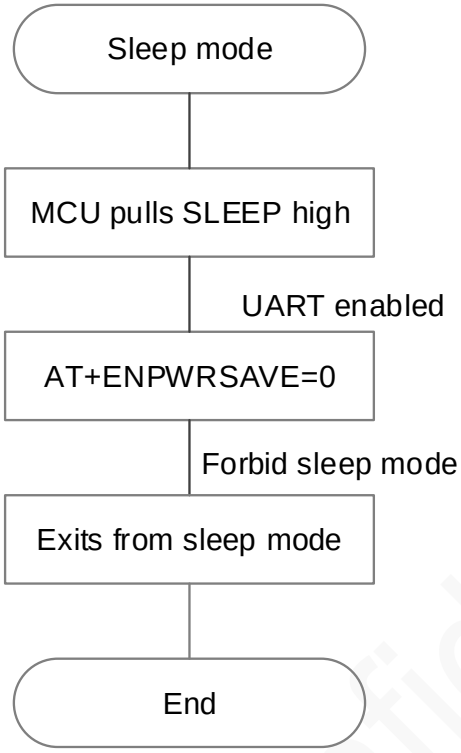


Figure 5-15 Process of processing data service in sleep mode



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Figure 5-16 Process of waking up the module

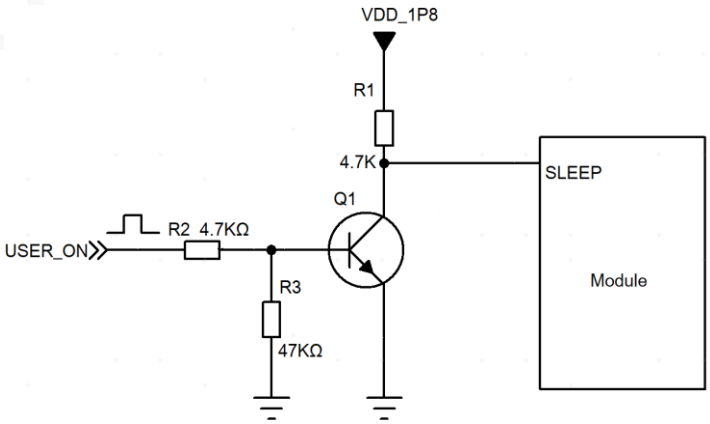


To wake up the module using the MAIN_DTR pin, it is recommended to design your circuit by referring to Figure 5-17.

Enter/exit sleep mode (send "AT+ENPWRSAVE=1" command to the serial interface of the module to enable sleep and wake-up function of the module.)

- Enter into sleep mode: keep USER_ON low
- Exit from sleep mode: keep USER_ON high

Figure 5-17 Reference design of module sleep and wake-up control



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5.3 Peripheral Interfaces

N706B offers a variety of peripheral interfaces.

Naming Conventions:

- The I/O direction indicated by the module peripheral interface pin name is in line with the module.
- The peripheral pin naming follows the peripheral component itself. For example, UART_TXD is the module‘s data send pin and MCU_RXD is the MCU’s receive. These pins should be interconnected.

In the process of MCU selection and design, it is important to note whether the pin signal naming is based on the module or the MCU.

5.3.1 USB

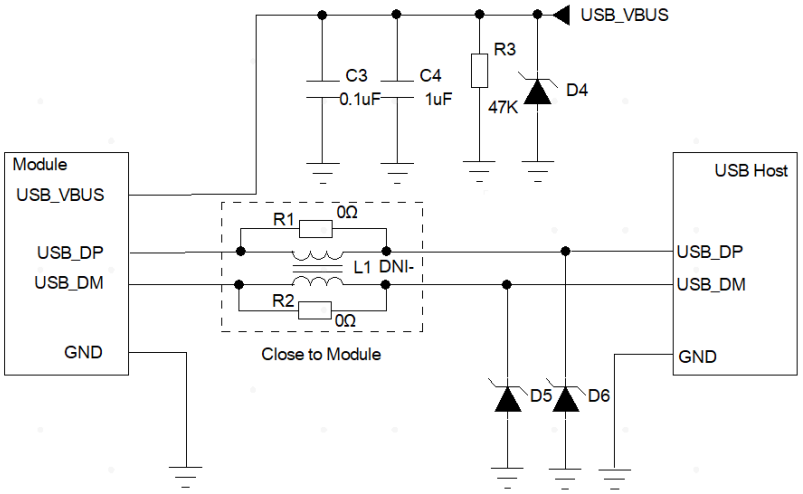
During software debugging, the USB interface is essential and must include test points in the design for testing and troubleshooting purposes.

Signal	Pin SN	I/O	Functional description	Remarks
USB_VBUS	61	PI	USB power supply	3.0 V < USB_VBUS < 5.25 V, typical value: 5 V.
USB_DP	59	AIO	USB data +	-
USB_DM	60	AIO	USB data -	-

N706B can implement program download, data communications, and debugging through the USB interface. Only slave mode is supported for USB of the module, which can be used as required. The recommended USB connection circuit is shown in Figure 5-18.

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Figure 5-18 Recommended design of the USB interface



Schematic Design Guidelines

- Place decoupling capacitors C1 (1μF) and C2 (0.1μF) on the USB_VBUS line for filtering. Add ESD protection devices to the power line, ensuring the reverse working voltage (V_{RWM}) exceeds 6V.
- It is recommended to connect a common-mode choke (L1) between the MCU and the module to suppress EMI interference. If L1 is used, resistors R1 and R2 should not be installed. If the USB is not connected to other devices and only test points are reserved, the common-mode choke can be omitted.
- To ensure the signal integrity of USB data lines, L1, R1, and R2 should be placed close to the module. Additionally, the resistors should be placed near each other, and the leads connecting to the test points should be kept as short as possible.
- Ensure that the junction capacitance of the ESD components (D2 and D3) in parallel to the USB_DP and USB_DM traces be less than 0.5 pF.
- Resistors of less than 10 Ω are needed in series at the USB_DM and USB_DP pins (R1, R2) to enhance ESD performance.

PCB Design Guidelines

- Place the filter capacitor on the USB_VBUS as close to the module pins as possible, and place the ESD component as close to the USB connector as possible.
- Place the paralleled ESD components (D2, D3) on the USE_DP and USB_DM traces as close to the USB connector as possible.
- It is important to route USB signal traces as differential pairs in the inner layer of the PCB with ground surrounding. The impedance of the USB differential traces should be 90 Ω. Shield the USB traces from other signal traces.
- USB traces should be routed away from crystal oscillators, oscillators, magnetic devices, and RF signals to avoid interference.

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5.3.2 UART

In environments with complex electromagnetic interference, the module's UART is susceptible to external electromagnetic interference, which can cause serious communication abnormalities. Therefore, please note the following when routing the UART signal trace on the PCB:

- Parallel a 5 to 10 pF capacitor near the module's UART_RXD and UART_TXD pins for filtering.
- Route UART traces far away from potential sources of interference.
- Route UART_RXD and UART_TXD traces on the inner layers of the PCB, maintain short trace lengths, avoid extensive routing on the top or bottom layers, and use a three-dimensional ground layout for these pins to enhance signal integrity.

Table 5-2 UART pin information

Interface type	Supported baud rate (bps)	Default baud rate (bps)	Description
MAIN_UART	4800, 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600	115200	Used for data transmission and AT command communication; supports RTS and CTS hardware flow control.
AUX_UART	115200	115200	Data transmission
DBG_UART	115200	115200	Outputs module logs

Table 5-3 UART pin definitions

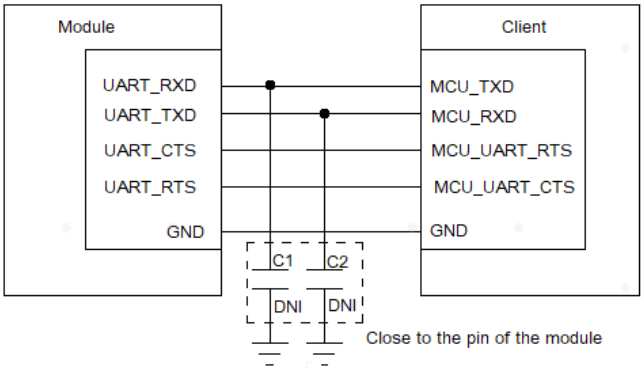
Signal	Pin SN	I/O	Functional description	Remarks
MAIN_RXD	17	DI	Main UART data input	Used for AT communications
MAIN_TXD	18	DO	Main UART data output	
MAIN_CTS	22	DI	Clear to send (main serial interface)	1.8 V voltage domain. Connected to the RTS of MCU. Leave this pin open if unused.
MAIN_RTS	23	DO	Request to send (main serial interface)	1.8 V voltage domain. Connected to the CTS of MCU. Do not repurpose as GPIO. Leave this pin open if unused.
AUX_RXD	28	DI	UART data input	AUX_RXD is used for data transmission (unavailable in sub-models with GNSS).
AUX_TXD	29	DO	UART data output	

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DBG_RXD	38	DI	UART data input	DBG_RXD is use only for debugging. Can be used to capture AP log. Test points must be reserved.
DBG_TXD	39	DO	UART data output	

N706B provides three UART interfaces. The UART supports a 1.8 V voltage and its default baud rate is 115200 bps. The following figure shows recommended design of the interface.

Figure 5-19 Reference design of the UART interface



Schematic Design Guidelines

- Ensure signal flow direction matches the connection orientation.
- Avoid using diodes for voltage-level translation.
- Use capacitors C1 and C2, typically 5 - 10 pF, adjusting as necessary based on testing.

Select one of three recommended circuits based on logic level and rate differences. Three voltage-level translation circuits are recommended based on the differences in logic levels and rates.

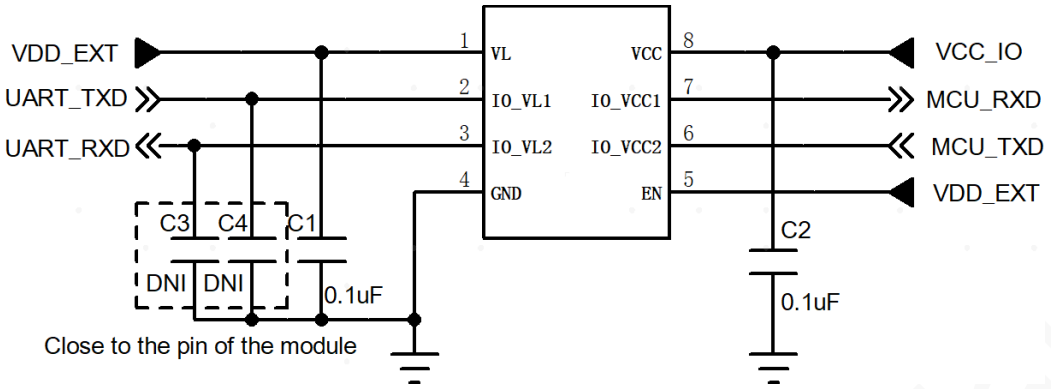
Adjust component values in these circuits based on actual test outcomes. Pay attention to the different types of voltage-level translation circuits.

- **If UART baud rate > 115200 bps**

  Refer to “Recommended voltage-level translation circuit 1”, as shown in Figure 5-20.

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Figure 5-20 Recommended voltage-level translation circuit 1



- VL: reference voltage for IO_VL1 and IO_VL2 (1.5 V - 5.5 V).
- VCC: reference voltage for IO_VCC1 and IO_VCC2 (1.5 V - 5.5 V).
- EN is the enable pin, which works at a voltage of greater than VL-0.2 V. In the above circuit, the EN pin is connected to VDD_EXT and the voltage translator is always working.
- It is recommended to reserve positions for filter capacitors C3 and C4 (with a recommended capacitance of 5 to 10 pF) and adjust based on actual test results.

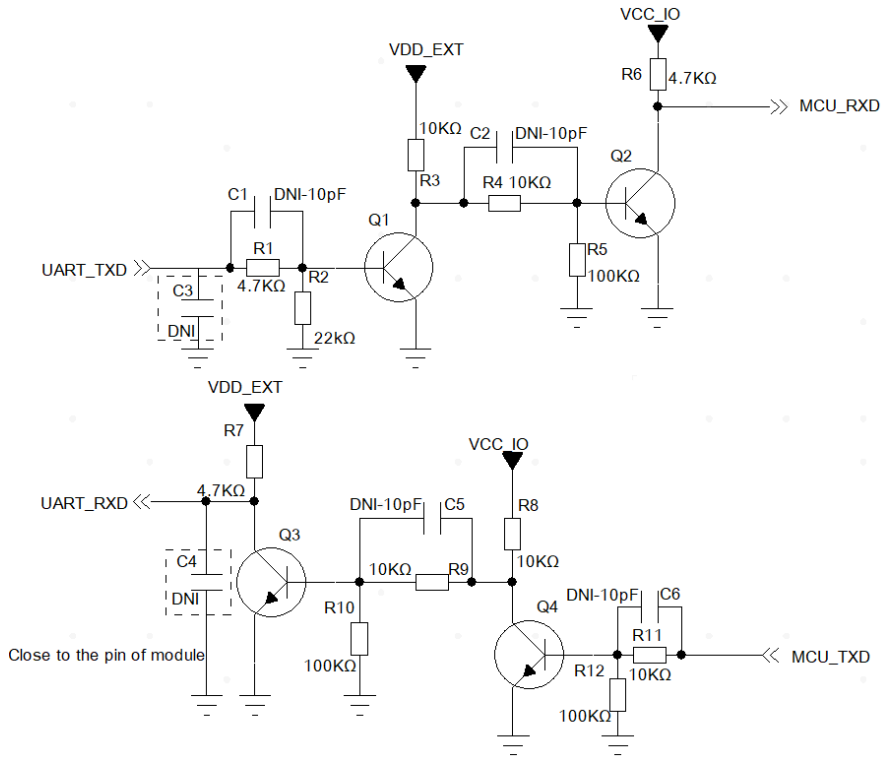
- **If UART baud rate ≤ 115200 bps**

  Designing the TXD and RXD serial interfaces by referring to recommended voltage-level translation circuit 2 and designing the CTS and RTS serial interfaces by referring to recommended voltage-level translation circuit 3 are recommended. As shown in Figure 5-21 and Figure 5-22.

Adjust component values in these circuits based on actual test outcomes. Pay attention to the different types of voltage-level translation circuits.

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Figure 5-21 Recommended voltage-level translation circuit 2



MCU_TXD and MCU_RXD: MCU's transmitting and receiving interfaces.

UART_TXD and UART_RXD: Module’s transmitting and receiving interfaces.

VCC_IO is the IO voltage of the MCU. VDD_EXT is the IO voltage of the module.

Schematic Design Guidelines

- Ensure the triode's base voltage operates within the specified temperature range for full conduction.
- Reserve acceleration capacitors (C1, C2, C5, C6) to potentially adjust the circuit's delay.
- It is recommended to reserve positions for filter capacitors C3 and C4 (with a recommended capacitance of 5 to 10 pF) and adjust based on actual test results.

- Single-Triode Voltage-Level Translation Circuit

  The circuit translates the voltage through the turn-on and turn-off of the triode, and dual triode can achieve higher pressure difference after translation.

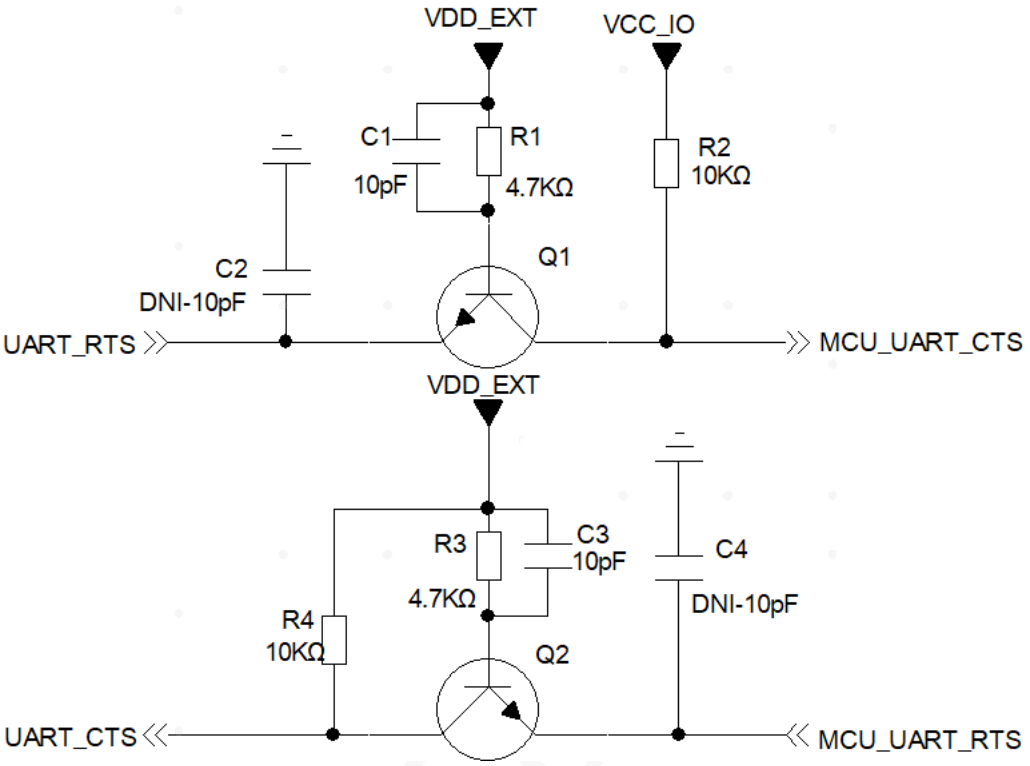
Schematic Design Guidelines

- Ensure the triode's base voltage operates within the specified temperature range for full conduction.

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- It is recommended to reserve the acceleration capacitor, which can adjust the delay of the voltage-level translation circuit in some cases.

Figure 5-22 Recommended voltage-level translation circuit 3



MCU_CTS and MCU_RTS are the MCU-side signals, while UART_CTS and UART_RTS are the module-side signals. VCC_IO is the IO voltage of the MCU. VDD_EXT is the IO voltage of the module.

The circuit translates the voltage-level through the turn-on and turn-off of the triode, and the translation is unidirectional. Pay attention to the signal flow.

Schematic Design Guidelines

- Maintain the voltage difference between high and low levels below 2 V.
- Use an accelerating capacitor and adjust it according to test results.
- The base voltage of the triode is the lower voltage of the two sides.

5.3.3 USIM

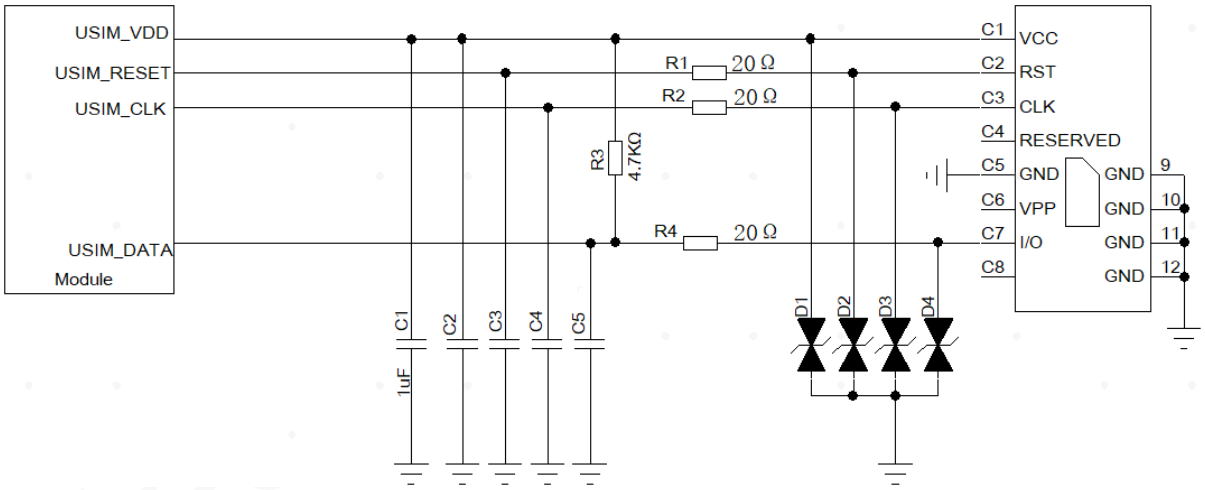
Signal	Pin SN	I/O	Functional description	Remarks
USIM1_DATA	11	B	USIM data	This pin is pulled up to USIM1_VDD via a 4.7 kΩ resistor.
USIM1_RESET	12	DO	USIM reset	-

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USIM1_CLK	13	DO	USIM clock	
USIM1_VDD	14	PO	USIM power supply	Both 1.8 V and 3.0 V SIM types are supported: activation and deactivation with an automatic voltage switch from 1.8 V to 3.0 V is implemented.
USIM2_DATA	64	B	USIM data	This pin is pulled up to USIM2_VDD via a 4.7 kΩ resistor.
USIM2_RESET	63	DO	USIM reset	-
USIM2_CLK	62	DO	USIM clock	-
USIM2_VDD	65	PO	USIM power supply	Only support 1.8V.

N706B provides 2 USIM interfaces: SIM1 supports 1.8V/3V auto-sensing, while SIM2 only supports 1.8V. The reference schematic diagram of the USIM card interface is shown in Figure 5-23.

Figure 5-23 Reference design for the USIM card interface



Schematic Design Guidelines

- USIM_VDD is the power supply of the USIM card, with load capacity of up to 200 mA. It is solely used for power supply for USIM card; do not power other loads. The pin must be reserved for a regulator capacitor and a filtering capacitor, with a recommended regulator capacitor (C1) of more than 1 uF and a recommended filtering capacitor (C2) of 0.1 uF.
- The USIM_DATA pin has an internal pull-up by default. It is recommended to reserve a 4.7 kΩ pull-up resistor to USIM_VCC.

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- USIM_CLK is the clock signal pin for USIM card. In applications with complex electromagnetic environments that have high requirements for ESD protection, it is recommended to add ESD protection diodes (junction capacitance ≤ 7pF) on each signal trace or add an integrated TVS array.
- Connect a resistor (≤ 20 Ω) to USIM_DATA, USIM_RESET, USIM_CLK, and USIM_DET (close to the card connector) in series to enhance ESD performance.
- C3-C5 are in parallel to each USIM signal trace and are used to mount high-frequency filter capacitors. Their capacitance value should be no more than 33 pF. Please adjust it according to the actual debugging results.



The module's USIM interface does not support hot-plugging by default. If you require this feature, the N706B module's pin79 can be used for SIM card detection, but enabling hot-plug detection on this pin will disable the SPI function. A 47kΩ resistor must be connected in series and pulled up to 1.8V. The software must support USIM hot-plug detection; please consult Neoway FAE for the specific software version.

PCB Design Guidelines

- USIM cards and their signal traces often be jammed by RF radiation, resulting in USIM abnormalities. The USIM interface should be kept away from switching power supplies, antenna areas, and RF circuit areas that may generate interference.
- The USIM card holder should be placed close to the module and try to ensure the routing length of the USIM card signal traces do not exceed 200 mm.
- On the USIM traces, connect the series resistor and ESD protection component close to the USIM card.
- Surround USIM traces with ground to improve EMC. To prevent USIM_CLK and USIM_DATA from mutually interfering, their routing should not be too close.

5.3.4 I2C

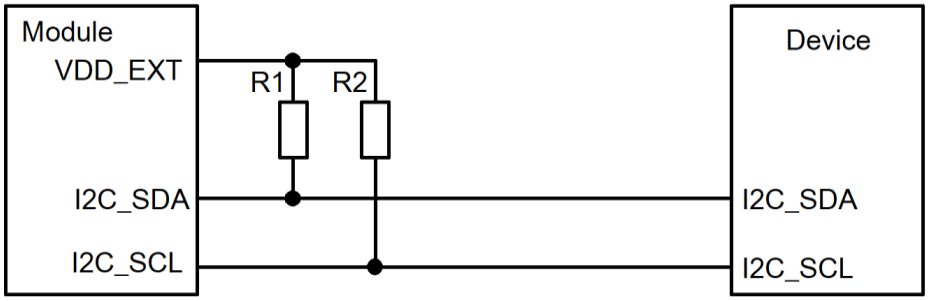
Signal	Pin SN	I/O	Functional description	Remarks
I2C1_SCL	57	DO	I2C clock	Connect this pin to VDD_EXT through a 4.7k pull-up resistor.
I2C1_SDA	58	B	I2C data	
I2C2_SDA	66	B	I2C data	
I2C2_SCL	67	DO	I2C clock	

N706B provides two I2C interfaces with 1.8 V level, supporting only the master mode. Their transmission rates in standard mode and fast mode are respectively 100 Kbps and 400 Kbps. For the pull-up resistors, select a proper resistance based on the actual debugging result; it is recommended

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to select the resistance ranging from 1.8 kΩ to 10 kΩ (default resistance 4.7 kΩ). Figure 5-24 shows the reference design of the I2C interface.

Figure 5-24 Reference design of I2C interface



5.4 Other Functional Interfaces

5.4.1 ADC

The module provides two 12-bit ADC channels, with measurable voltage range 0.1 V - 1.2 V, which can be used for temperature measurement and other related measurement. For details, see the *Neoway_N706B_AT_Commands_Manual*.

Signal	Pin SN	I/O	Functional description	Remarks
ADC2	9	AI	General A/D interface	-
ADC1	96	AI	General A/D interface	-

Schematic Design Guidelines

- When designing the ADC interface, a voltage divider circuit must be reserved. The resistor connected to the power supply in each voltage divider should have a resistance value between 100 kΩ and 1 MΩ.
- The precision of the divider resistors affects the ADC sampling error. It is recommended to use resistors with 1% precision. For higher ADC accuracy, resistors with 0.5% precision are suggested.
- Note that ESD protection devices should be added when using this pin.

PCB Design Guidelines

- To improve the voltage measurement accuracy of the ADC interface, it is recommended to apply ground shielding around the traces.

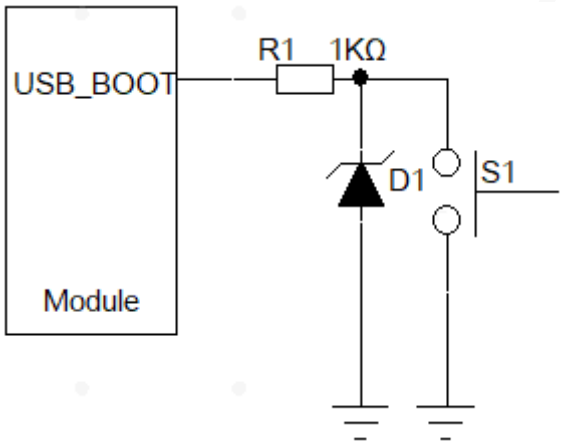
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5.4.2 USB_BOOT

Signal	Pin SN	I/O	Functional description	Remarks
USB_BOOT	82	B	Emergency download control	-

The USB_BOOT pin is designed for enabling emergency download mode. To enter emergency download mode you should connect USB_BOOT of the module to ground and then power up the module. This is the last method to handle issues that result in startup or running failures. To facilitate the subsequent software upgrade and debugging of the product, it is recommended to reserve this pin. The following figure shows the reference design of this pin. Adding an ESD component to protect USB_BOOT in the circuit is required.

Figure 5-25 Reference design for the emergency download pin



5.4.3 NET_STATUS

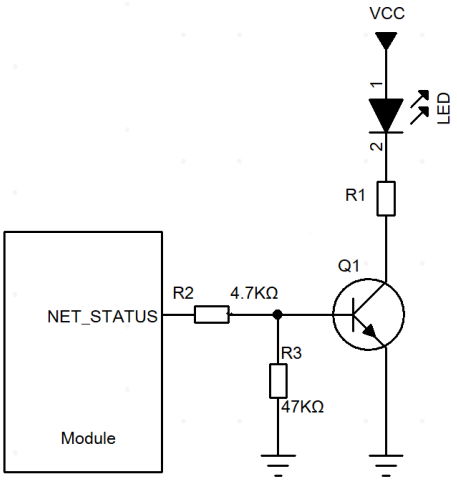
Signal	Pin SN	I/O	Functional description	Remarks
NET_STATUS	16	DO	Network status indication	Leave this pin open if unused.

NET_STATUS is the network status indicator pin of the module. It outputs PMW waves of duty cycle varying with the status of the module and drives an LED indicator to blink at different modulation frequencies. You can send AT commands to enable the LED indicator to blink in different states. For details, see *Neoway_N706B_AT_Commands_Manual*.

The NET_STATUS indication pin outputs a high level of 1.8 V. Do not use this pin to directly drive LED indicator. It is recommended to drive LED indicators through a triode.

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Figure 5-26 Driving LED indicators with a triode



5.4.4 STATUS

Signal	Pin SN	I/O	Functional description	Remarks
STATUS	25	DO	Operating status indication	Leave this pin open if unused.

STATUS is a network status indicator pin of the module. It outputs PMW waves of duty cycle varying with the status of the module and drives LED indicators to blink at different modulation frequencies. You can send AT commands to enable the LED indicator to blink in different states. For details, see *Neoway_N706B_AT_Commands_Manual*.

The STATUS indication pin outputs a high level of 1.8 V, and do not use this pin to directly drive LED indicator. It is recommended to drive LED indicators by controlling a triode. Figure 5-26 shows the recommended reference design.

5.4.5 RING

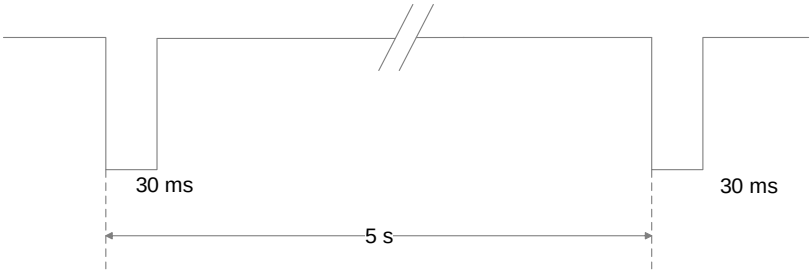
Signal	Pin SN	I/O	Functional description	Remarks
MAIN_RI	20	DO	Incoming call & SMS indicator control	The duration of the low pulse can be configured via software.

- Incoming call RING indication

  When there is a voice call, during the ringing process, the RING pin outputs a low pulse with a period of 5s and a pulse width of 30ms. The duration of the low pulse can be configured by software. After the call is answered, the pin restores to a high level, as shown in the following figure.

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Figure 5-27 Incoming call RING indication



- SMS RING indication

  When an SMS message arrives, the RING pin will generate a low pulse as a notification. As shown in the following figure.

Figure 5-28 SMS RING indication



5.5 RF Interfaces

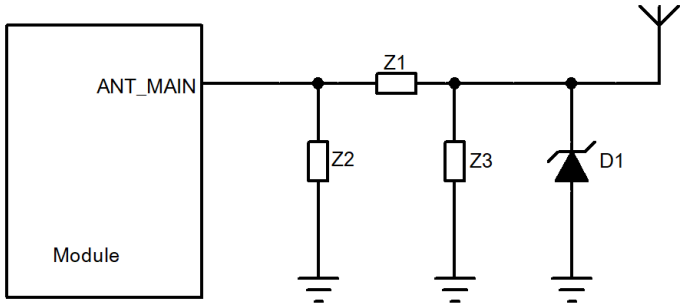
Signal	Pin SN	I/O	Functional description
ANT_WIFI_SCAN	98	AI	WIFI_SCAN antenna
ANT_MAIN	35	AIO	Main antenna
ANT_GNSS	2	AIO	GNSS antenna

5.5.1 ANT_MAIN Antenna Interface

For the module to be applicable to your PCB, the characteristic impedance of the antenna interface should be controlled at 50 Ω and the impedance of the trace from the module's antenna interface to the antenna needs to be kept at 50 Ω to allow reception of radio frequency (RF) signals. In the circuit design, a matching network is essential for antenna matching in the circuit design. The π-type matching circuitry is preferred, as shown in the figure below.

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Figure 5-29 π-type RF matching schematic



Schematic Design Guidelines

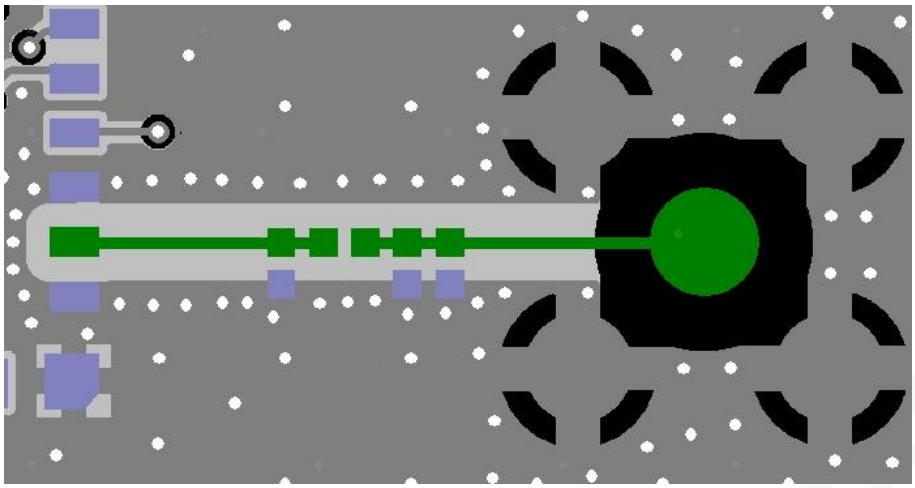
- The components in the RF matching circuit consist of capacitors, inductors, and 0 Ω resistors. Place these RLC components as close to the antenna interface as possible.
- If there is static electricity introduced on the antenna, it is recommended to add electrostatic protection. You can use ESD devices with ultra-low junction capacitance, such as TVS diodes with a junction capacitance of less than 0.5pF. Additionally, ensure that the V_{RWM} of the TVS diode is above 15V.

PCB Design Guidelines

- Surround RF traces with grounded copper foil. Drill as many via holes as possible on the copper and control RF trace impedance to 50 Ω.
- Keep the PCB routing between the RF pin and the antenna interface under 50 Ω impedance control, and make the trace length as short as possible.
- To reduce the potential impact on antenna performance caused by the significant parasitic capacitance from the large RF pad in the SMA RF seat, the entire bottom area of the SMA RF seat's RF signal is hollowed out. Additionally, the ground pad next to the module's RF pad is not treated with a thermal pad but should be in complete contact with the ground, as illustrated in the figure below.

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Figure 5-30 Recommended RF PCB design



- A reasonable distance should be kept between ANT_MAIN and ANT_WIFI_SCAN to avoid mutual interference that may affect reception performance.
- On the PCB, keep the RF signals and components far away from digital circuits, switching power supplies, power transformers, power inductors, clock signals, etc.

5.5.2 ANT_WIFI_SCAN Antenna Interface

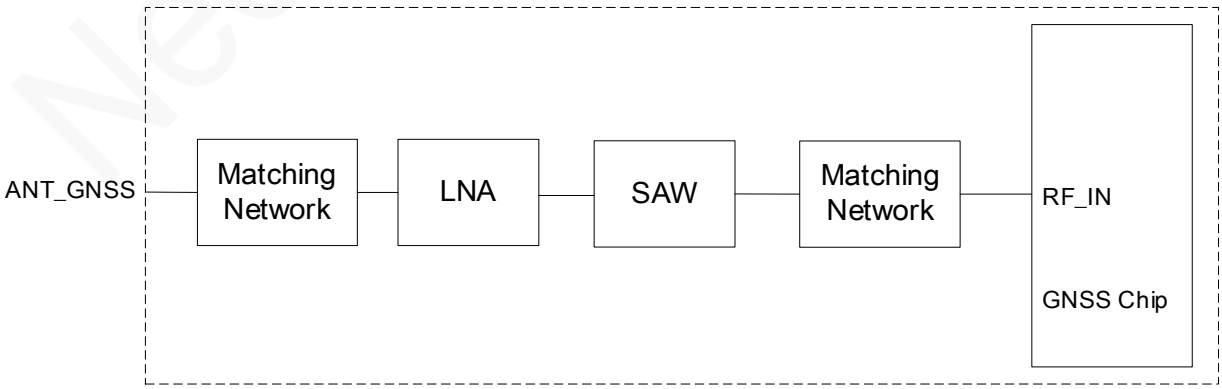
The impedance characteristic of the N706B module's WIFI_SCAN RF interface must be 50Ω.

For schematic and PCB design of the WIFI_SCAN antenna interface, refer to the requirements detailed in section 5.5.1 "ANT_MAIN Antenna Interface".

5.5.3 ANT_GNSS

The GNSS RF interface of the N706B module also requires a 50Ω impedance characteristic. The internal structure of the GNSS RF connection within the module is shown in the diagram below.

Figure 5-31 GNSS RF structure

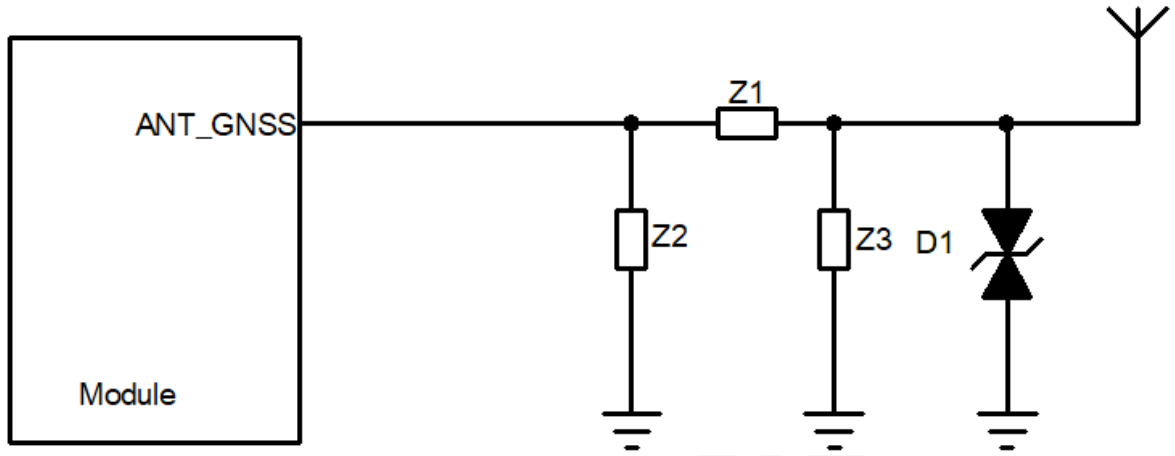


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Reference design of passive GNSS antenna

After the GNSS antenna receives the GNSS satellite signal, it is transmitted to the ANT_GNSS pin of the N706B module through the PCB traces. The following figure shows a reference design of the passive GNSS antenna.

Figure 5-32 Reference design of passive GNSS antenna



PCB Design Guidelines

- For cautions of PCB design between GNSS interface and antenna, refer to the PCB design guidelines in section 5.5.1 “ANT_MAIN Antenna Interface”.
- Keep the GNSS RF circuit away from the main/ WIFI_SCAN antenna circuits as far as possible on your application PCB. Otherwise, the RF performance may be affected.

Reference design of active GNSS antenna

It is strongly recommended to use active antennas to achieve better performance.

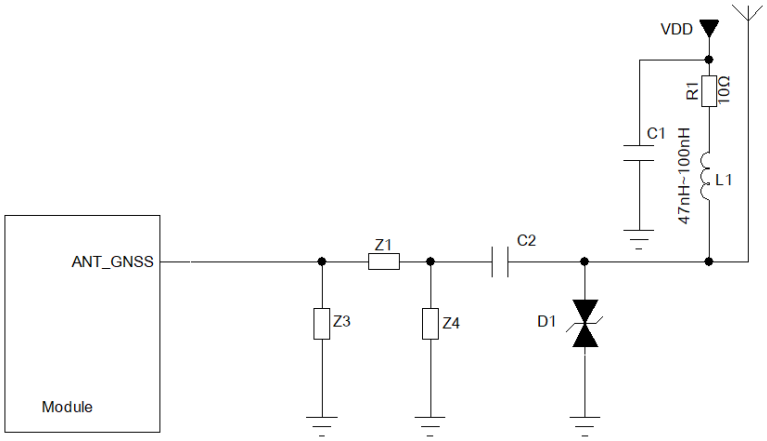
The GNSS chip has a maximum gain limit for the external receiving chain. Exceeding this limit can degrade GNSS performance or even damage the chip. If the module has a built-in LNA, the gain requirements for the active antenna are as follows:

N706B-CN: Active antenna gain should not exceed 28 dB, with a recommended gain range of 16-19 dB. Noise figure should be ≤ 3 dB.

After the GNSS antenna receives GNSS satellite signals, the signals are amplified by the front LNA (Low Noise Amplifier) inside the active antenna and then sent to the ANT_GNSS pin of the module through the feeder and PCB traces. The following figure shows the reference design of the active GNSS antenna.

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Figure 5-33 Reference design of active GNSS antenna



Schematic Design Guidelines

- Pay attention to the gain limit of the active antenna and the recommended range of the active antenna.
- Common active antennas are powered by 3.3V~5V. Although the active antenna itself has low power consumption, it requires a low-noise LDO (recommended noise level below 40 μV_{RMS}) to supply power to the antenna via an inductor of 47nH-100nH, as shown in Figure 5-33. It is recommended to use the module's GPIO to control the PWR_EN of the LDO output.
- R1 is a current-limiting resistor to prevent excessive LDO output current in case of an active antenna short circuit, which could lower system power voltage and affect normal operation. The value of R1 should be determined based on the active antenna current specifications. If using an LDO with overcurrent protection, R1 can be omitted.
- For ESD protection, use ultra-low junction capacitance ESD devices, with junction capacitance recommended to be less than 0.5 pF.

PCB Design Guidelines

- For cautions of PCB design between GNSS interface and antenna, refer to the PCB design guidelines in section 5.5.1 “ANT_MAIN Antenna Interface”.
- 50 Ω impedance control is required for both antenna feeders and the PCB traces between the antenna interface and LNA, and the traces should not be too long. The power supply of the active antenna is fed via an inductor of 47 nH - 100 nH from the signal line of the antenna.
- Keep the GNSS RF circuit away from the main/ WIFI_SCAN antenna circuits as far as possible on your application PCB. Otherwise, the RF performance may be affected.

5.5.4 Antenna Assembly

The antenna used by the module must comply with mobile device standards, with a standing wave ratio between 1.1 and 1.5, an input impedance of 50 Ω, and varying gain requirements based on the environment. You can choose an appropriate antenna according to specific application scenarios and

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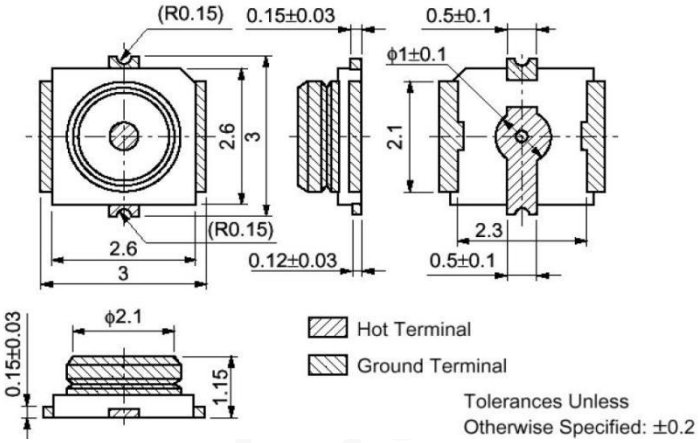
environments.

The module antenna interface can connect to rubber rod, sucker, or internal PIFA antennas. Ensure proper shielding between external antennas and RF pins. For RF cable connections, keep the external RF cables away from all interference sources, especially digital signals and switching power supplies.

The following methods are commonly used to assemble antenna:

- Connecting to a GSC RF connector
  Recommended model MM9329-2700RA1. The following figure shows its encapsulation specifications. The connection to the external antenna can be made through RF lines.

Figure 5-34 Murata RF connector encapsulation specifications



- Connecting to an external antenna by soldering
  It may result in stability, consistency, and RF performance issues and is not recommended.

The following show the two types of connections.

Figure 5-35 RF cable connections



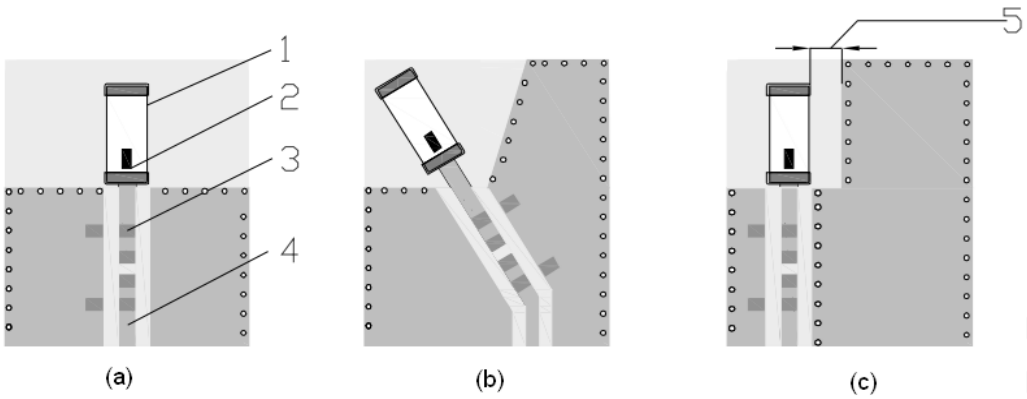
- PCB antenna or chip antenna

  The module works in a wide frequency range, but it is difficult for PCB antennas or ceramic antennas to cover a wide frequency. Therefore, this connection method is recommended only for 2.4 GHz Wi-Fi or BT/ BLE antennas.

The following figure shows the layout of the 2.4 GHz ceramic chip antenna. SLDA52-2R540G-S1TF is used as an example.

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Figure 5-36 Antenna layout

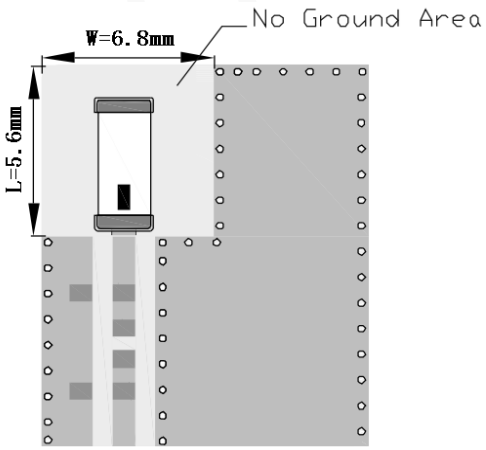


If there is enough space on the PCB, refer to Figure 5-36 (a).

- 1 Chip antenna
- 2 Feeding mark
- 3 Solder pad of the matching circuitry
- 4 50 Ω characteristic impedance RF trace

In Figure 5-36, the position marked as ”5“ depicts the area between the antenna and the ground. As shown in Figure 5-37, the "No Ground Area".

Figure 5-37 Clearance area (No Ground Area) around the antenna on the PCB board



For more details, please refer to the antenna manual and instruction documentation.

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6 Electrical Characteristics and Reliability

This chapter describes the electrical characteristics and reliability of the N706B module, including the input and output voltage and current of the power supply, current consumption of the module in different states, operating and storage temperature range, and ESD protection characteristics.

6.1 Electrical Characteristics

If the voltage is lower than the threshold, the module might fail to start. If the voltage is higher than threshold or there is a voltage burst during the startup, the module might be damaged permanently.

Table 6-1 N706B operating conditions

Parameter		Minimum value	Typical value	Max. value	Remarks
VBAT	V_{in}	3.4 V	3.8 V	4.3 V	-
	I_{in}	N/A	N/A	2 A	The above data was obtained using the Neoway EVB in a lab. It was measured while connected to a CMW500, with the module operating at its maximum transmission power and a VBAT of 3.8V.

Table 6-2 N706B-CN current consumption (typical)

State / Network standard and band	Sleep (mA)	Idle (mA)	Active(mA)
FDD-LTE: B1, B3, B5, B8	<1.5	<13.5	＜600
TDD-LTE: B34, B39, B40, B41	<1.5	<15	＜300

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Table 6-3 N706B-EA current consumption (typical)

State / Network standard and band	Sleep (mA)	Idle (mA)	Active (mA)
FDD-LTE: B1, B3, B5, B7, B8, B20, B28	<1.5	<13.5	<700

Table 6-4 N706B-LA current consumption (typical)

State / Network standard and band	Sleep (mA)	Idle (mA)	Active (mA)
FDD-LTE: B1, B2, B3, B4, B5, B7, B8, B28, B66	<1.5	<13.5	<700

Table 6-5 N706B-CN GNSS current (Typical)

Description	Condition	Current (mA)
Cold start (AT+CFUN=0)	Instrument	<41
Tracking mode (AT+CFUN=0)	Instrument	<48

All test data in this section are based on Neoway EVK and measured in a laboratory environment using a CMW500.

6.2 Temperature Characteristics

Parameter	Minimum value	Typical value	Max. value
Operating	-30℃	25℃	75℃
Extended	-40℃	25℃	85℃*
Storage	-40℃	25℃	90℃

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In the event that the temperature in the operating environment surpasses the designated thresholds, certain RF performance indicators of the module may deteriorate and fall short of the 3GPP specification requirements. However, despite this, the module's regular functionality will not be significantly affected. It is possible for the RF performance to return to the required 3GPP standards once the temperature is normalized.

6.3 ESD Protection Characteristics

Since electronic products need to undergo strict ESD testing, the module pins need to have ESD protection capabilities. You need to add corresponding ESD protection according to the application industry of the product when designing related products to ensure product quality.

Test environment: humidity 45%; temperature 25 ℃; using N706-CB-Mini-EVB

Table 6-6 ESD Protection Characteristics

Contact point	Contact discharge	Air discharge
GND	±8 kV	±15 kV
ANT	±4 kV	±8 kV
Shielding cover	±8 kV	±15 kV

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7 RF Characteristics

The module provides connectivity on FDD-LTE (Cat 1) and TDD-LTE (Cat 1) networks. This chapter details the wireless RF characteristics.

7.1 Operating Frequency Bands

Table 7-1 N706B operating frequency bands

Operating band	Uplink	Downlink
FDD-LTE B1	1920 - 1980 MHz	2110 - 2170 MHz
FDD-LTE B2	1850 - 1910 MHz	1930 - 1990 MHz
FDD-LTE B3	1710 - 1785 MHz	1805 - 1880 MHz
FDD-LTE B4	1710 - 1755 MHz	1710 - 1755 MHz
FDD-LTE B5	824 - 849 MHz	869 - 894 MHz
FDD-LTE B7	2500 - 2570 MHz	2620 - 2690 MHz
FDD-LTE B8	880 - 915 MHz	925 - 960 MHz
FDD-LTE B20	832 - 862 MHz	791 - 821 MHz
FDD-LTE B28	703 - 748 MHz	758 - 803 MHz
FDD-LTE B66	1710 - 1780 MHz	2110 - 2200 MHz
TDD-LTE B34	2010 - 2025 MHz	2010 - 2025 MHz
TDD-LTE B39	1880 - 1920 MHz	1880 - 1920 MHz
TDD-LTE B40	2300 - 2400 MHz	2300 - 2400 MHz
TDD-LTE B41	2496 - 2690 MHz	2496 - 2690 MHz

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7.2 TX Power and RX Sensitivity

Table 7-2 N706B RF transmit power

Band	Max power	Min. power
FDD-LTE B1	23 dBm±2 dB	< -40 dBm
FDD-LTE B2	23 dBm±2 dB	< -40 dBm
FDD-LTE B3	23 dBm±2 dB	< -40 dBm
FDD-LTE B4	23 dBm±2 dB	< -40 dBm
FDD-LTE B5	23 dBm±2 dB	< -40 dBm
FDD-LTE B7	23 dBm±2 dB	< -40 dBm
FDD-LTE B8	23 dBm±2 dB	< -40 dBm
FDD-LTE B20	23 dBm±2 dB	< -40 dBm
FDD-LTE B28	23 dBm±2 dB	< -40 dBm
FDD-LTE B66	23 dBm±2 dB	< -40 dBm
TDD-LTE B34	23 dBm±2 dB	< -40 dBm
TDD-LTE B39	23 dBm±2 dB	< -40 dBm
TDD-LTE B40	23 dBm±2 dB	< -40 dBm
TDD-LTE B41	23 dBm±2 dB	< -40 dBm

Table 7-3 N706B RF RX sensitivity

Band	Reception sensitivity
FDD-LTE B1	< -98dBm
FDD-LTE B2	< -98dBm
FDD-LTE B3	< -98dBm
FDD-LTE B4	< -98dBm
FDD-LTE B5	< -97dBm
FDD-LTE B7	< -98dBm
FDD-LTE B8	< -97dBm
FDD-LTE B20	< -98dBm
FDD-LTE B28	< -98dBm
FDD-LTE B66	< -98dBm
TDD-LTE B34	< -99dBm

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TDD-LTE B39	< -99dBm
TDD-LTE B40	< -99dBm
TDD-LTE B41	< -99dBm

The preceding indicators are tested in a shielded environment in a laboratory. The LTE band indicators are the test results when the bandwidth is 10 MHz, the modulation mode is QPST and RB is set according to the protocol. On no-shielded environments, deviations may exist in the receiver sensitivity of some individual bands due to the interference.

7.3 GNSS Technical Parameters

Table 7-4 GNSS technical parameters

Parameter	Description
GPS L1 operating frequency	1575.42±1.023 MHz
GLONASS operating frequency	1597.5 - 1605.9 MHz
BDS operating frequency	1559.1 - 1563.1 MHz
Tracking sensitivity	≤-159 dBm
Acquisition sensitivity	≤145 dBm
Positioning precision (open space)	<2m (room temperature)
Hot start (open space)	≤3s (room temperature)
Cold start (open space)	≤30s (room temperature)
Update frequency	1 Hz - 10 Hz
Max. positioning speed	500 m/s
Max. positioning acceleration	≤4G
Max. positioning altitude	18000m
GNSS data type	NMEA-0183, ICOE Protocol
GNSS antenna type	Passive/active antenna

Tracking sensitivity and acquisition sensitivity are the results of signaling test on GSS7000. The values are the maximum values obtained from multiple measurements performed on samples. The module's hot start time refers to the time taken for the GNSS to restart when the module is powered on.

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7.4 WIFI_SCAN Technical Parameters

Table 7-5 WIFI_SCAN technical parameters

Operating bands	Transmitting rate
802.11b (2.4G)	1/2/5.5/11 Mbps

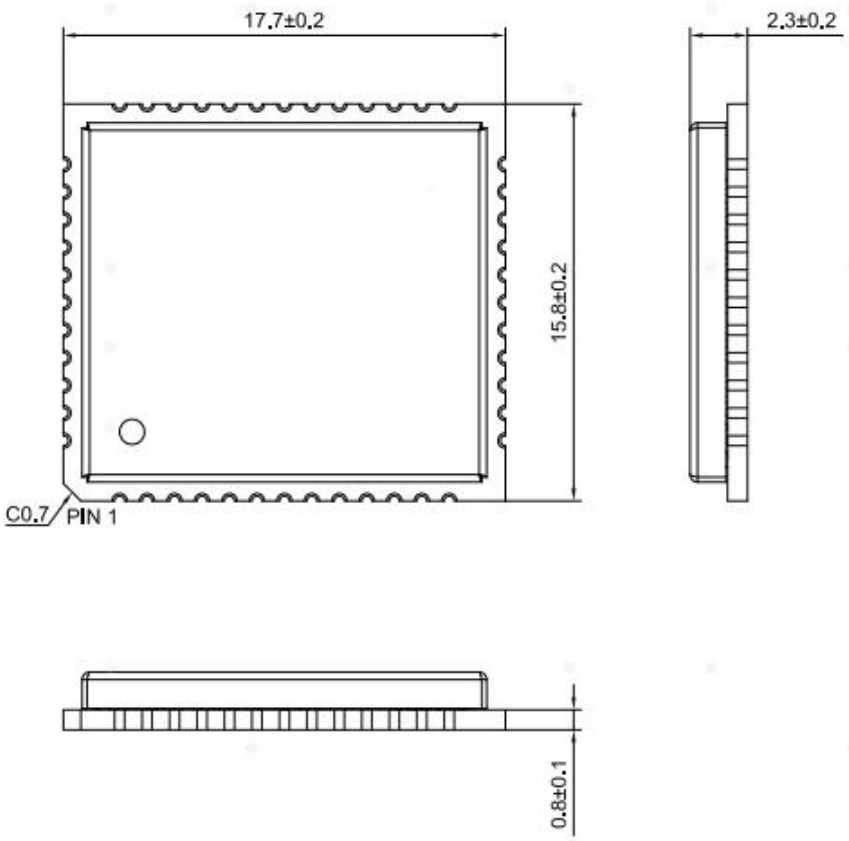
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8 Mechanical Characteristics

This chapter describes mechanical characteristics of the N706B module.

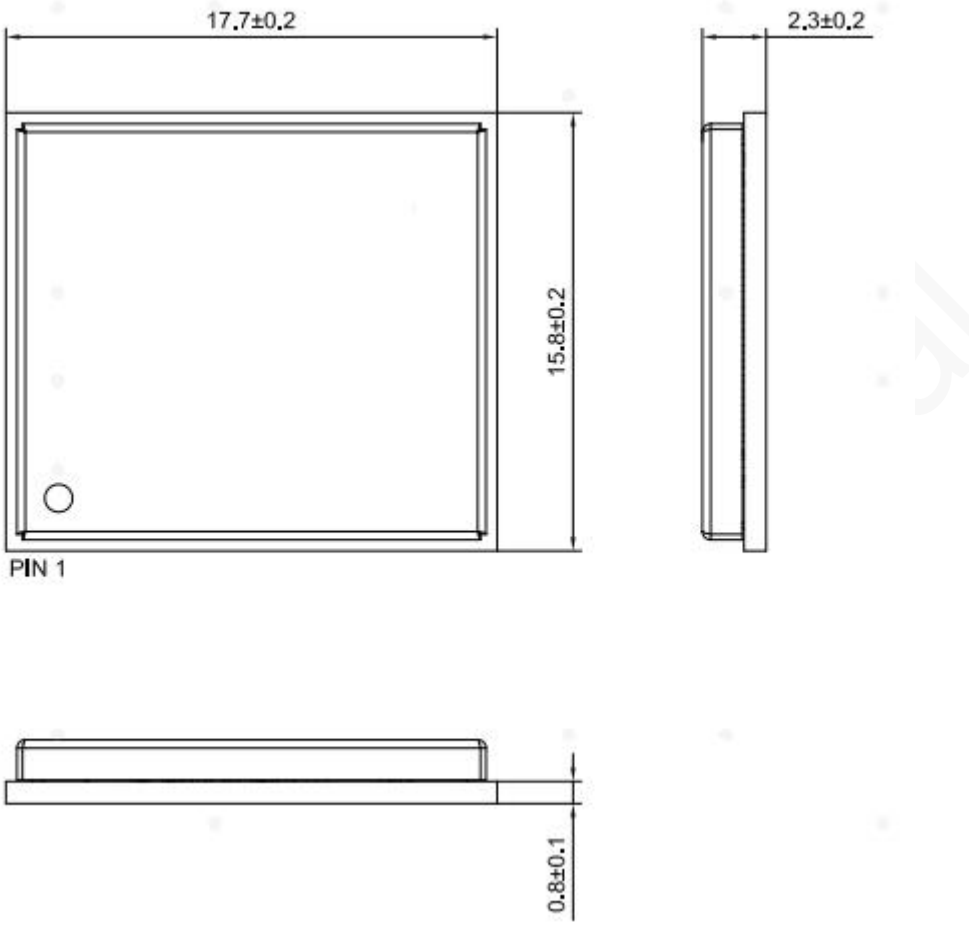
8.1 Dimensions

Figure 8-1 N706B-CN top and side view dimensions (unit: mm)



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Figure 8-2 N706B-EA/LA top and side view dimensions (unit: mm)



8.2 Labeling

The N706B label is laser etched, and can withstand a high temperature of 260°C.

Figure 8-3 N706B-CN labels



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Figure 8-4 N706B-EA label



Figure 8-5 N706B-LA label



The above figure is for reference only. For authentic appearance, please refer to the module that you receive from Neoway.

8.3 Packaging

Due to the use of surface-mount technology (SMT) for solder reflow in the N706B modules, moisture-proof packaging methods are employed to prevent moisture absorption from production to customer use. These methods include aluminum foil bags, desiccants, humidity indicator cards, reels, and vacuum sealing. Such measures ensure the products remain dry, thereby extending their usability.

8.3.1 Tape&Reel

Neoway modules are delivered as reeled tapes as shown below:

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8.3.2 Moisture Sensitivity Level

N706B is a level 3 moisture-sensitive electronic element, in compliance IPC/ JEDEC J-STD-020 standard.

- Sealed storage life: 12 months at temperatures below 30℃ and humidity less than 60%.
- Production environment condition: 30°C/60%
- The module should be pre-baked under the following circumstances:
  The vacuum-sealed packaging is removed for more than 168 hours.
- When the humidity card comes with the package shows above 10%, it indicates that the package has been exposed to high humidity levels.

After unpacking, avoid leaving the module exposed to air for too long. If exposed to air for a long time, the module could become damp, increasing the risk of damage during reflow soldering or laboratory soldering. Bake it before mounting the module. The baking conditions depend on the degree of moisture absorption, but it is recommended to bake at no less than 120℃/6 hours. Do not bake Neoway modules while contained in a tape and rolled up in reels. For baking, place modules individually onto the oven tray.

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9 Mounting N706B onto Application Board

This chapter describes the package of N706B, the recommended footprint of the application PCB, and SMT specifications.

9.1 N706B Module Footprint

Figure 9-1 Bottom view of N706B-CN PCB package (unit: mm)

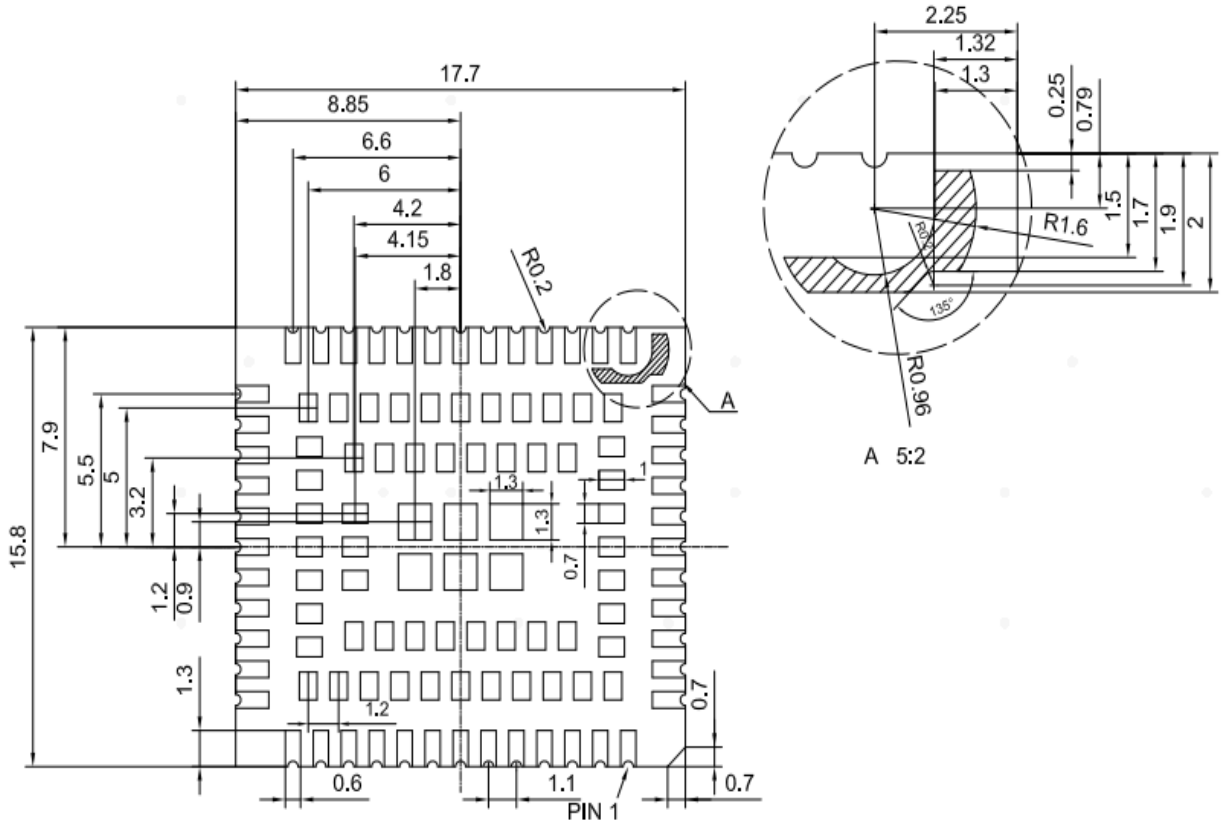
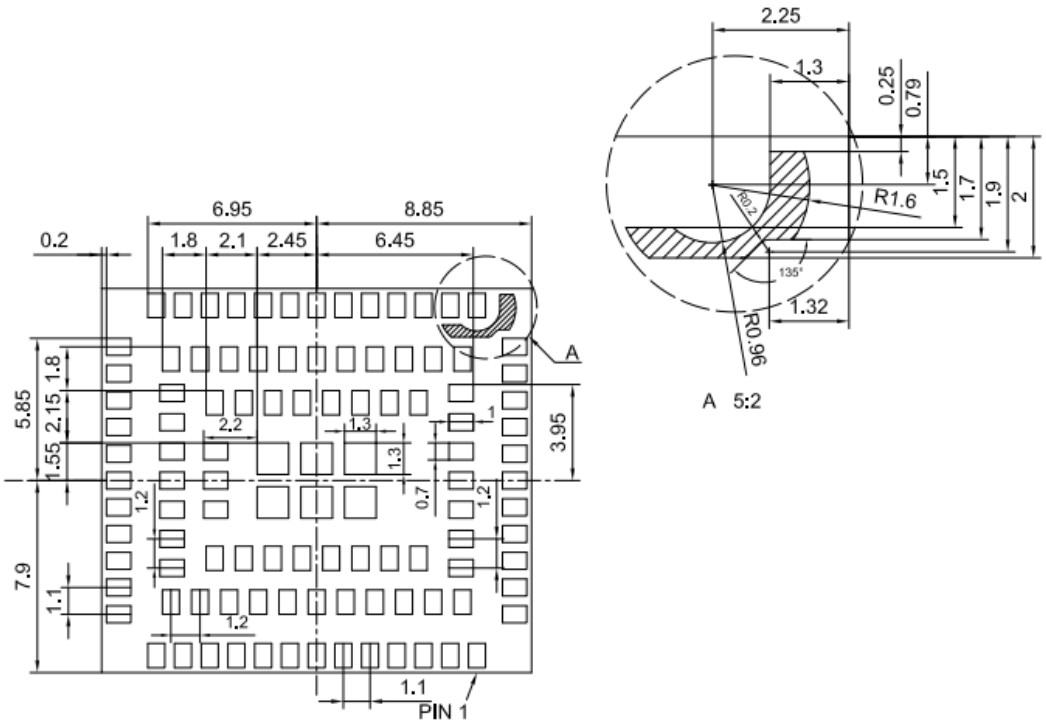


Figure 9-2 Bottom view of N706B-EA/LA PCB package (unit: mm)



9.2 Application PCB Package

The N706B module consists of a total of 109 pins. The N706B-CN uses an LCC+LGA (109-pin) package, while the N706B-EA / LA use an LGA (109-pin) package. The recommended PCB footprint is as follows.

- To ensure proper operation of the module, only GND via-holes and copper pouring are allowed within the shaded area "" of the PCB package.
- To achieve higher yield during module production, it is recommended that the distance between other components on the PCB board and the module edge be at least 3 mm to avoid the risk of tin connection when using stepped stencil.

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Figure 9-3 N706B-CN module PCB foot print (top view, unit: mm)

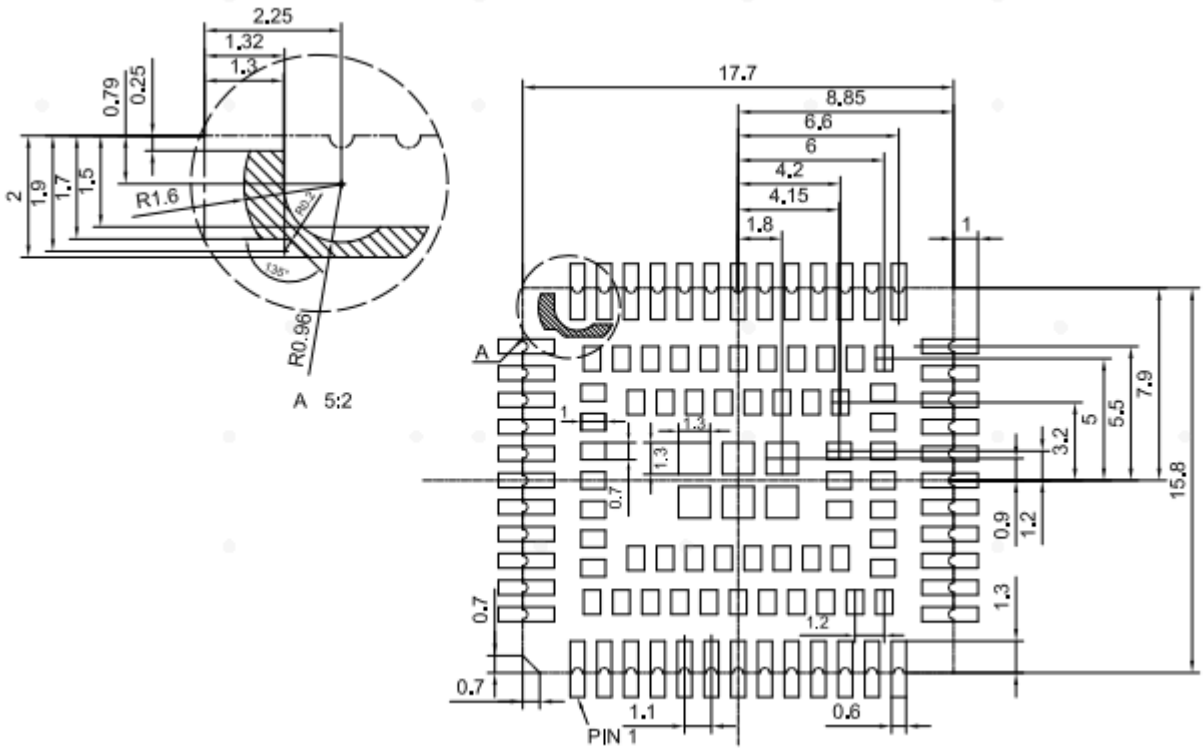
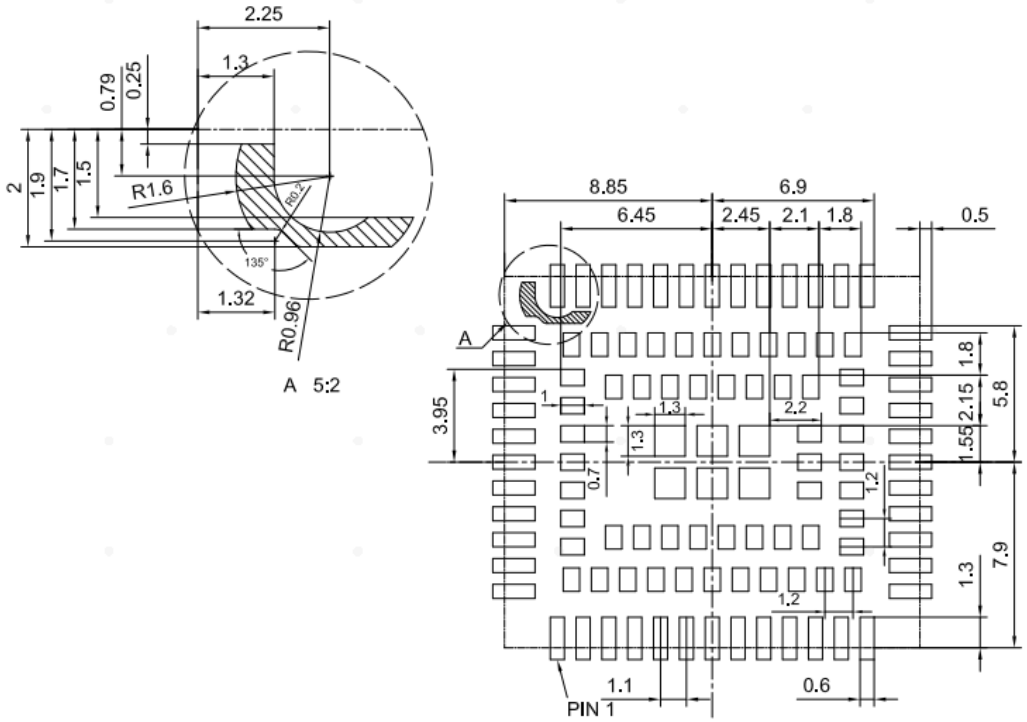


Figure 9-4 N706B-EA/LA module PCB foot print (top view, unit: mm)



9.3 Stencil

The recommended stencil thickness is at least 0.15 mm to 0.20 mm, which can be fine-tuned according to the actual situation, preferably 0.18 mm. For details, see *Neoway_Communication_Modules_Mounting_Guide*.

9.4 Solder Paste

The thickness of solder paste and the flatness of PCB are essential for the production yield.

Neoway modules use lead-free solder paste technology. It is recommended to use the same type of solder paste as used in our module production technique.

Users are generally not recommended to use leaded solder paste due to the following reason:

Leaded solder paste has a melting point 35°C lower than lead-free, resulting in lower temperatures and shorter times in reflow processes, which may cause LCC/LGA components in the module to be in a semi-melted state during the second reflow, leading to cold solder joints.

When using solder pastes with lead, ensure the reflow temperature exceeds 220°C for over 45 seconds, with a peak temperature of 240°C.

9.5 Oven Temperature Profile

Neoway will not provide warranties for temperature-sensitive element abnormalities caused by improper temperature control.

Different products may have varying reflow temperature profiles. Before reflow soldering, it is essential to conduct reflow temperature testing using the temperature profile board of the specific product. If a reflow fixture was used during production, it must also be used during the temperature testing to prevent heat absorption by the fixture, which could cause cold solder joints on the module. Only after passing the test can the product proceed to reflow production.

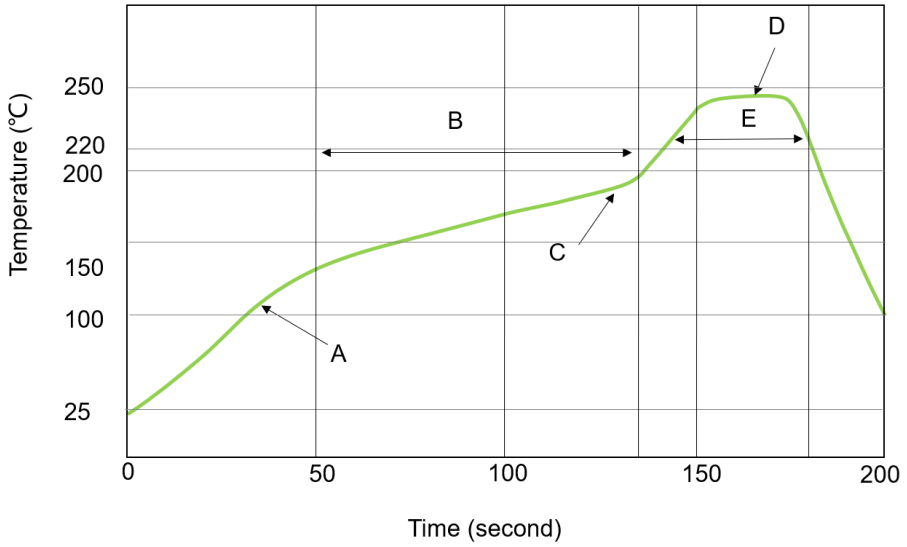
The following reflow temperature profiles are provided for your reference and can be optimized based on actual solder paste usage and other factors.

During production, no organic solvents (such as alcohol, isopropanol, acetone, trichloroethylene, etc.) should be used to wipe the module labels or shielding covers to avoid discoloration of the labels and covers.

Thin or long PCB might bend during SMT. So, use loading tools during the SMT and reflow soldering process to avoid poor solder joint caused by PCB bending.

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Figure 9-5 Oven temperature profile



When testing reflow oven temperature, attach a test point to the solder pad at the bottom of the module to ensure it reaches the soldering temperature. The recommended lead-free reflow temperature profile is shown in the image above, with specific parameters as listed below. Adjust and optimize according to the technical requirements of the solder paste used.

Technical parameters:

- Ramp up rate: 1 to 3°C/sec
- Ramp down rate: -5 to -1°C/sec
- Soaking zone: 150 - 180°C, Time: 60 - 100s
- Reflow zone: >220°C, Time: 40 - 80s
- Peak temperature: 235 – 245℃

For information about cautions in storage and mounting, refer to *Neoway_Communication_Modules_Mounting_Guide*.

When removing the module manually from your application PCB board is required, use heat guns with great opening, adjust the temperature to about 350°C (depending on the type of the solder paste), and heat the module till the solder paste is melted. Then remove the module using tweezers. Do not shake the module at high temperatures while removing it. Otherwise, the components inside the module might get misplaced.

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A Abbreviations

Abbreviation	Full name
ADC	Analog-to-Digital Converter
AI	Analog Input
AO	Analog Output
AIO	Analog Input/Output
ARM	Advanced RISC Machine
BT	Bluetooth
bps	Bits per Second
CCC	China Compulsory Certification
CEP	Circular Error Probable
CNR	Carrier to Noise Rate
CS	Chip Select
CTS	Clear to Send
DC	Direct Current
DCS	Digital Cellular System
DI	Digital Input
DIO	Digital Input/Output
DL	Downlink
DO	Digital Output
DPSK	Differential Phase Shift Keying
DQPSK	Differential Quadrature Phase Shift Keying
DRX	Discontinuous Reception
DTR	Data Terminal Ready
ECM	Ethernet Control Model
eDRX	Extended DRX
EGSM	Enhanced GSM
ESD	Electronic Static Discharge
ESR	Equivalent Series Resistance

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EVK	Evaluation Kit
FDD	Frequency Division Duplexing
FPC	Flexible Printed Circuit
FTP	File Transfer Protocol
GFSK	Gauss Frequency Shift Keying
GLONASS	GLOBAL NAVIGATION SATELLITE SYSTEM
GNSS	Global Navigation Satellite System
GPIO	General Purpose Input Output
3GPP	3rd Generation Partnership Project
GPRS	General Packet Radio Service
GPS	Global Positioning System
GSM	Global System for Mobile Communications
I2C	Inter-Integrated Circuit
IO	Input/Output
ISP	Image Signal Processor
LCC	Leadless Chip Carriers
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LGA	Land Grid Array
LTE	Long Term Evolution
MCLK	Main Clock
MCU	Microcontroller Unit
MIC	Microphone
PCB	Printed Circuit Board
PCS	Personal Communications Service
PWM	Pulse Width Modulation
QVGA	Quarter Video Graphics Array
RAM	Random Access Memory
RF	Radio Frequency
ROM	Read-only Memory
RTC	Real Time Clock
SD	Secure Digital
SDIO	Secure Digital Input Output

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SPK	Speaker
SPI	Serial Peripheral Interface
TDD	Time Division Duplex
UART	Universal Asynchronous Receiver-Transmitter
UL	Uplink
USB	Universal Serial Bus
USIM	Universal Subscriber Identity Module
VBAT	Battery Voltage
VSWR	Voltage Standing Wave Ratio
Wi-Fi	Wireless Fidelity
WCDMA	Wide-band Code Division Multiple Access
WCI	Wireless Coexistence Interface
WLAN	Wireless Local Area Network

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